

First International Computer, Inc

Protable Computer Group HW Department

Board name : Mother Board Schematic

Project : LM7W

Version : 0.2

Initial Date : August 31 , 2005

1. Schematic Page Description :
2. PCI & IRQ & DMA Description :
3. Block Diagram :
4. Nat name Description :
5. Board Stack up Description :
6. Schematic modify Item and History :
7. power on & off & S3 Sequence :
8. Layout Guideline :
9. switch setting

Manager Sign by:

Drawing by : Tom_Lin

Total confirm by: CC_TSAO

LAN Circuit check by:

Audio Circuit check by:

		First International Computer, Inc. 2F/L NO.300, Yang Guang St., Neihu 114 TAIPEI, TAIWAN, R.O.C. (886-2)8751-8751	
File LM7W < VIA VN800 + VT8237R >			
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1. Schematic Page Description :

LM7 Schematic Ver : 0.1

1. Title

2. Schematic Page Description

3. Block Diagram

4. ANNOTATIONS

5. Schematic Modify

6. Timing Diagram

7. DDR Layout Guideline

8. Banias Celeron(1/2)

9. Banias Celeron(2/2)

10. POWER (CPU CORE)

11. Thermal / VR_PWRGD / RTC

12. Clock Generator

13. Clock Buffer

14. VN800 (1/4)

15. VN800 (2/4)

16. VN800 (3/4)

17. VN800 (4/4)

18. DDR SO-DIMM1

19. DDR SO-DIMM0

20. VT1631 LVDS Transmitter

21. LCD Connector

22. CRT Connector
23. VT8237 (1/3)

24. VT8237 (2/3)

25. VT8237 (3/3)

26. Power Good & Fan Controller

27. CB1410 CardBus Controller

28. CB1410 CardBus Power SW./CNN

29. MINI PCI

30. VT6103L PHY

31. USB CNN

32. S-ATA HD / CD-ROM CNN

33. LPC PMU08

34. LPC KBC M3885X

35. INT KBC / GP Connector

36. MDC Connector

37. DIP Switch & LED

38. Firm Ware Hub / LID Switch

39. Reset Circuit

40. OVP / SCREW

41. ALC655 Audio Codec

42. G1432+1410 Audio Amplifier

43. H.P. Out / Audio CNN

44. DDR PWR
45. 1.5VDDA/S , 1.8/2.5VDDM/A

46. 3VDDM / 5VDDM

47. VCCP/1.5VDDM

48. 3VDDA / 5VDDA / PMU3/5V

49. POW-ON Controller

50. ADIN / Battery CNN

51. Charge Circuit / DCIN

52. Inverter Controller

53. Audio board

54. Switch transfer board

55. Update list

2. PCI & IRQ & DMA Description :

IDSEL	CHIP
AD17	Mini PCI(Wireless LAN)
AD23	CardBus

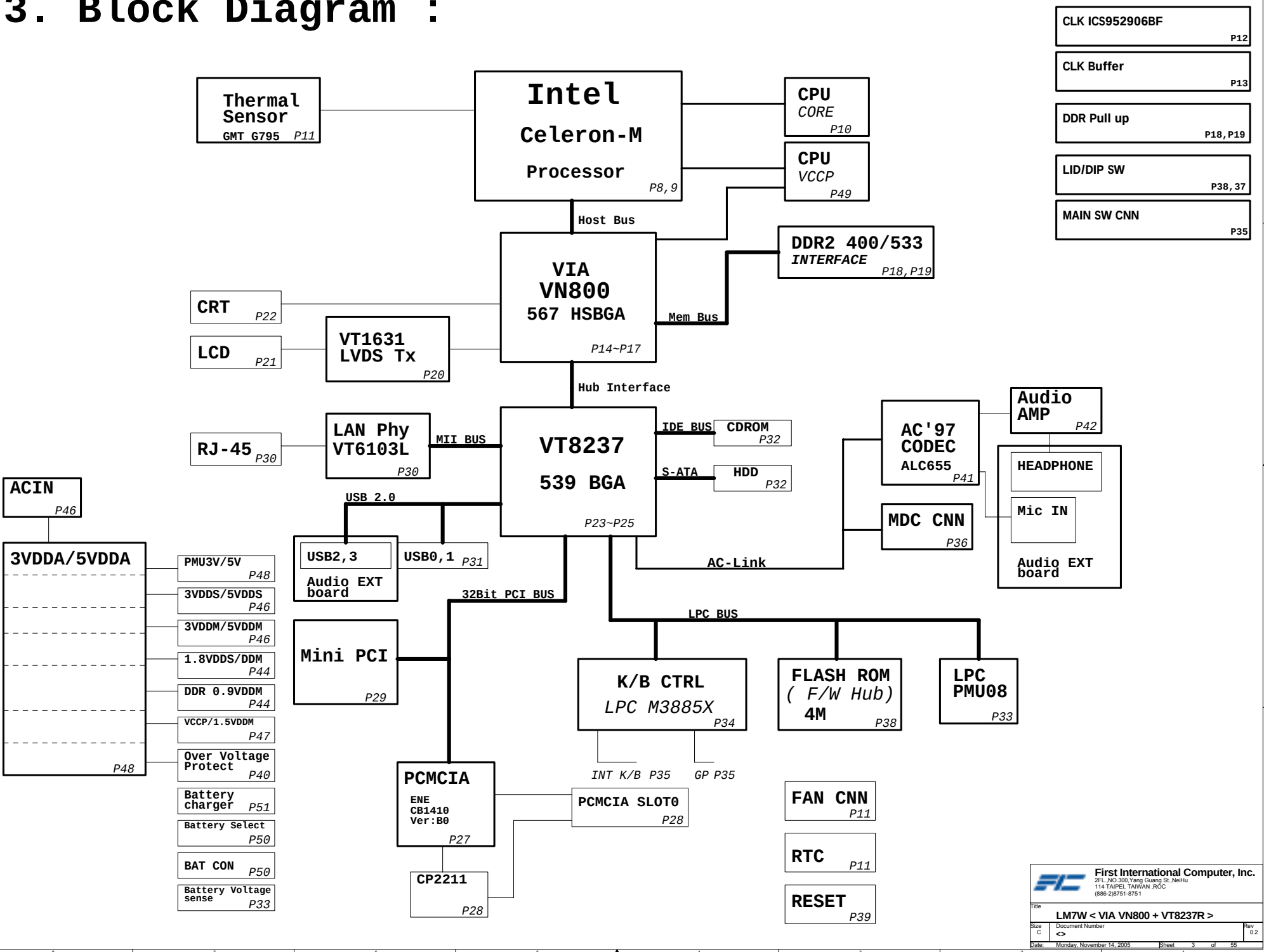
PCIINT	CHIP
IRQA	MiniPCI/NB
IRQB	MiniPCI/CardBus
IRQC	MiniPCI
IRQD	

BUSMASTER	CHIP
REQ	
REQ0 / GNT0	MiniPCI
REQ1 / GNT1	CardBus
REQ2 / GNT2	Mini PCI(Wireless LAN)
REQ3 / GNT3	
REQ4 / GNT4	

IRQ Channel	Description
IRQ0	System timer
IRQ1	Keyboard
IRQ2	(Cascade)
IRQ3	LAN / MODEM
IRQ4	Serial Port
IRQ5	AUDIO / VGA / USB
IRQ6	FLOPPY DISK
IRQ7	LPT
IRQ8	RTC
IRQ9	ACPI
IRQ10	FIR (Disable by default) (MODEM/LAN)
IRQ11	Cardbus
IRQ12	PS/2 mouse
IRQ13	FPU
IRQ14	HDD
IRQ15	CDROM

DMA Channel	Device
DMA0	FIR (disable by default) (MODEM / LAN)
DMA1	ECF
DMA2	FLOPPY DISK
DMA3	AUDIO
DMA4	(Cascade)
DMA5	Unused
DMA6	Unused
DMA7	Unused

3. Block Diagram :



4. Nat name Description :

Voltage Rails

DCIN	Primary DC system power supply
PMU5V	5.0V always on power rail by LATCH or ACIN
PMU3V	3.3V always on power rail by LATCH or ACIN
5VDDA	5.0V always on power rail by DCON or PSUSC0
3VDDA	3.3V always on power rail by DCON or PSUSC0
3VDDS	3.3V power rail
5VDDS	5.0V power rail
3VDDM	3.3V switched power rail
5VDDM	5.0V switched power rail
Vcore_CPU	Core Voltage for CPU

VCCP	1.05V for AGTL+ Termination Voltage
1.8VDDM	1.8V for CPU PLL Voltage
DDR_0.9VDDM	0.9V DDR Termination Voltage
1.5VDDM	1.5V switched power rail
1.5VDDS	1.5V power rail
1.5VDDA	1.5V always on power rail
2.5VDDS	2.5V power rail for DDR

Part Naming Conventions

C	= Capacitor
CN	= Connector
D	= Diode
F	= Fuse
L	= Inductor
Q	= Transistor
R	= Resistor
RP	= Resistor Pack
U	= Arbitrary Logic Device
Y	= Crystal and Osc

Net Name Suffix

0	= Active Low signal
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Signal Conditioning

D	= Damped (by a resistor)
Q	= Isolated (by a Q-switch)
L	= Filtered (by an inductor or bead)

5.Board Stack up Description

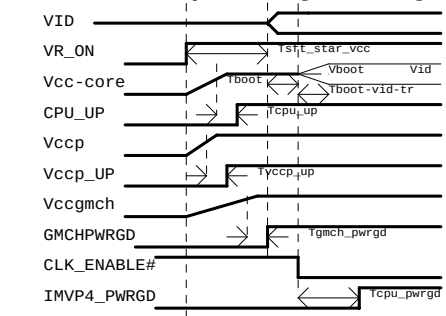
PCB Layers

Layer 1		Component Side, Microstrip signal Layer
Layer 2		Ground Plane
Layer 3		Stripline Layer (AGTL, CLOCK, DDR)
Layer 4		Power Plane
Layer 5		Ground Plane
Layer 6		Stripline Layer (Analog, LVDS, other)
Layer 7		Ground Plane
Layer 8		Solder Side, Microstrip signal Layer

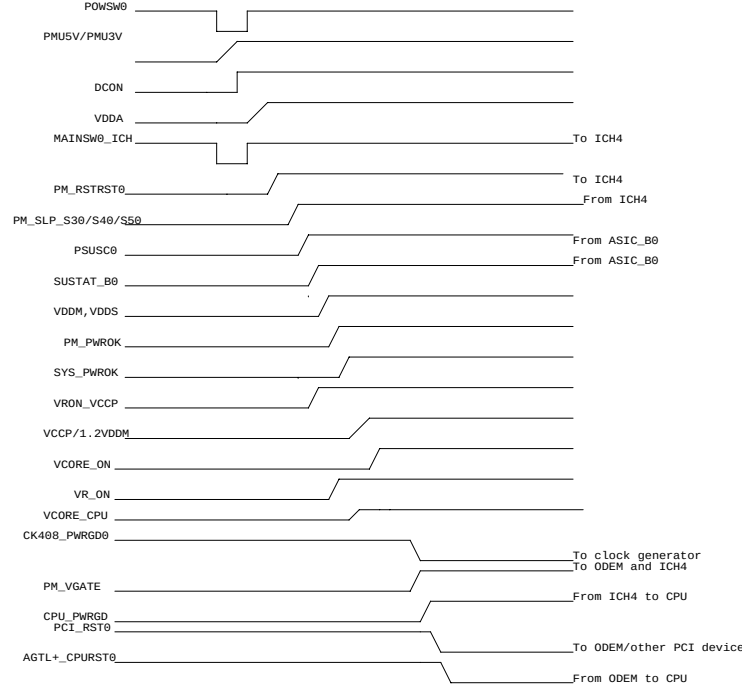
6.Schematic modify Item and History :

7. power on & off & S3 Sequence :

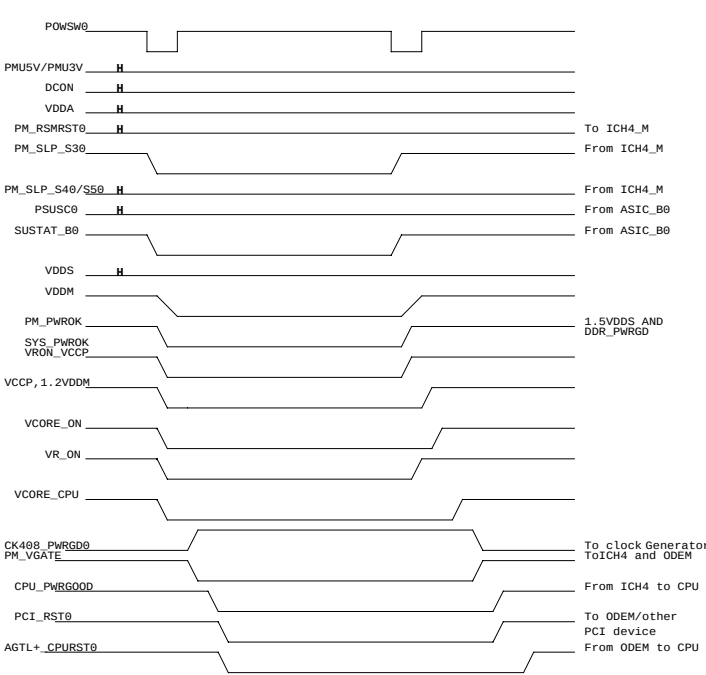
Power On Sequencing Timing Diagram



BATTERY ONLY POWER ON TIMING



S3 SUSPEND AND RESUME TIMING



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2/F, NO. 300, Yang Guang St., Neihu
114 TAIPEI, TAIWAN, R.O.C.
(886-2)8751-8751

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8. Layout Guideline :

Montara-GM DDR Layout Guidelines

Note that all length matching formulas are based on GMCH die-pad to SO-DIMM pin total length

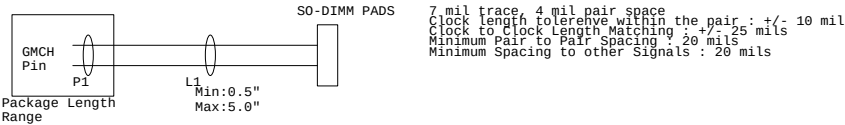
DDR Signal Groups

Group	Signal Name
Clocks	SCK[5:0] SCK# [5:0]
Data	SDQ[71:0] SDQS[8:0] SDM[8:0]
Control	SCKE[3:0] SCSE[3:0]
Command	SMA[12,6,3:0] SRAS# SCAS# SWE#
CPC	SMA[5,4,2,1] SMAB[5,4,2,1]
Feedback	RCVENOUT# RCVENIN#

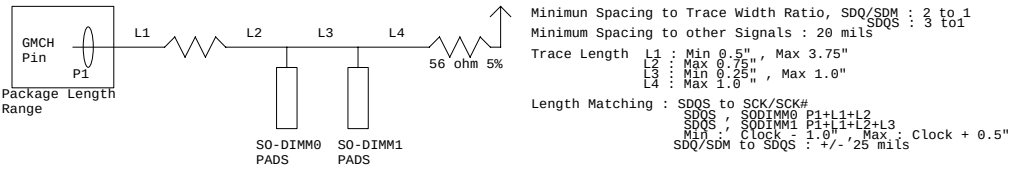
Length Matching Formulas

Signal Group	Minimum Length	Maximum Length
Control to Clock	Clock - 1.0"	Clock + 0.5"
Command to Clock	Clock - 1.0"	Clock + 2.0"
CPC to Clock	Clock - 1.0"	Clock + 0.5"
Strobe to Clock	Clock - 1.0"	Clock + 0.5"
Data to Strobe	Strobe - 25 mils	Strobe + 25 mils

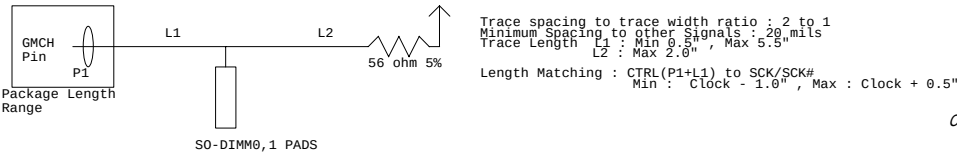
Clock Signals Topologies and Routing Guidelines



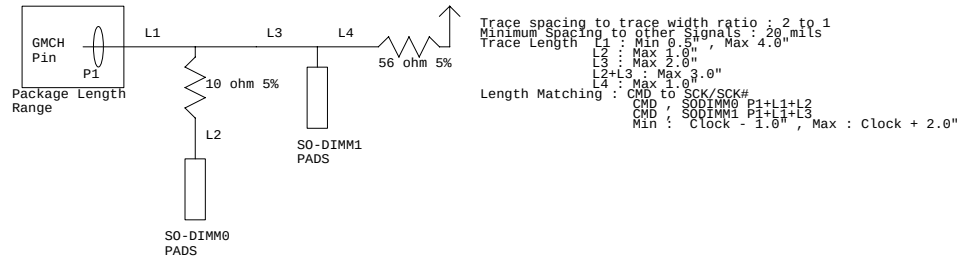
Data Signals Topologies and Routing Guidelines



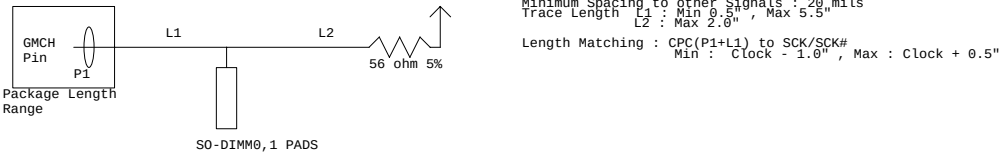
Control Signals Topologies and Routing Guidelines



Command Signals Topologies and Routing Guidelines



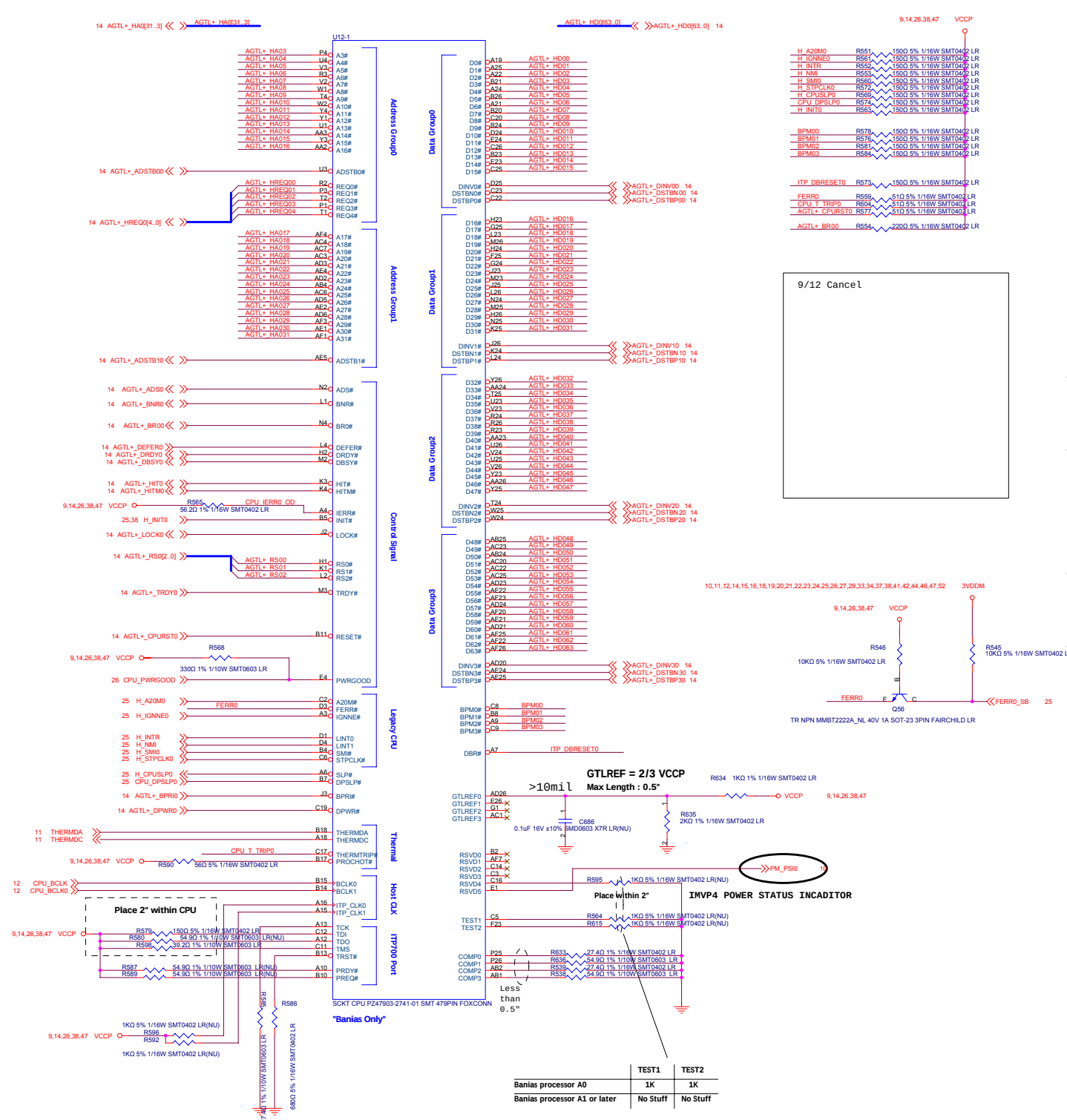
CPC Signals Topologies and Routing Guidelines



CLOCKS	LENGTH	TRACE / SPACE	NOTES
HCLKCPU[1..0] HCLKNB[1..0] HCLKITP[1..0]	2" ~ 8"	5 / 20 mils (5 mil space between + & -)	1. Differentials pairs with the same length (within 10 mil) 2. CPU & NB trace mismatch within 450 mil
66MCLK_ICH 66MCLK_GMCH AGPCLK_ATI	4.5" ~ 9.0" MAX : 8.5"	5 / 20 mils	66MCLK_ICH & AGPCLK_GMCH AGPCLK_ATI Length mismatch within 100 mils
PCLKICH PCLKCB PCLK1394 PCLKUSB20 PCLKOP PCLKFWH PCLKSIO PCLKLAN	4.5"-9.0"	5 / 20 mils	1. Making PCI length with minimum various 2. Max skew = 1ns
14MCLK_SIO 14MCLK_ICH 14MCLK_AC97	4.5"-9.0"	5 / 10 mils	
48MCLK_ICH 48MCLK_CB	3.5" ~ 12.5"	5 / 20 mils	

SDQ/SDM to SDQS Mapping

Signal	Mask	Relative To	Mismatching
SDQ[7..0]	SDM[0]	SDQS[0]	+/- 25 mil
SDQ[15..8]	SDM[1]	SDQS[1]	+/- 25 mil
SDQ[23..16]	SDM[2]	SDQS[2]	+/- 25 mil
SDQ[31..24]	SDM[3]	SDQS[3]	+/- 25 mil
SDQ[39..32]	SDM[4]	SDQS[4]	+/- 25 mil
SDQ[56..40]	SDM[5]	SDQS[5]	+/- 25 mil
SDQ[55..48]	SDM[6]	SDQS[6]	+/- 25 mil
SDQ[63..56]	SDM[7]	SDQS[7]	+/- 25 mil
SDQ[71..64]	SDM[8]	SDQS[8]	+/- 25 mil



System Bus Common Clock Signal Layout Guide :

ADSW# , BNR#, BPRI#, BRO#, DBSY#, DEFER#, DPWR#, DRDY#, HIT#, HITM#, LOCK#, RS[2..0]#, TRDY#, RESET#			
Transmission Line Type	Total Trace Length	Normal Impedance	Spacing (mils)
Strip-line(Int. Layer) Micro-strip(Ext. Layer)	1.0 ~ 6.5 inch	55+/-10%	4 & 8(Int. Layer) 5 & 10(Ext. Layer)

Source Synchronous DATA :

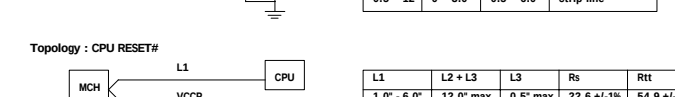
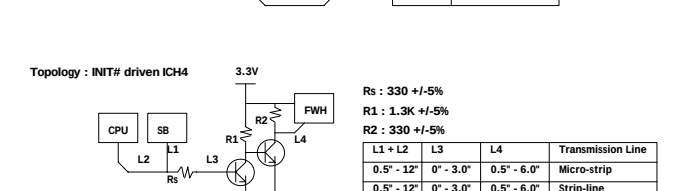
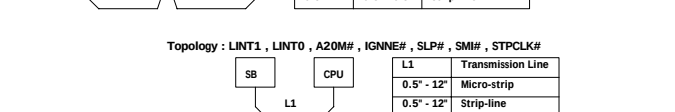
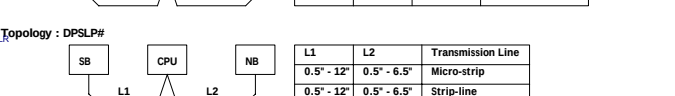
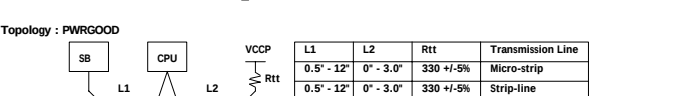
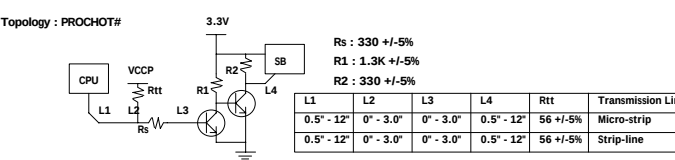
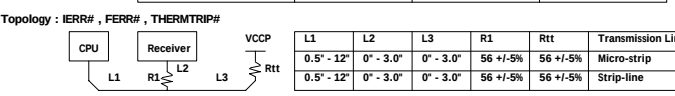
DATA#[63..0], DINV#[3..0], DSTBN#[3..0], DSTBP#[3..0]			
Transmission Line Type	Total Trace Length	Normal Impedance	Spacing (mils)
Strip-line	1.0 ~ 6.5 inch	55+/-10%	4 & 12

Signals Name	Signals Matching	Strobes associated with the group	Strobe Matching
DATA#[15..0], DINV#	+/- 100 mils	DSTBP0#, DSTBN0#	+/- 25 mils
DATA#[31..16], DINV1#	+/- 100 mils	DSTBP1#, DSTBN1#	+/- 25 mils
DATA#[47..32], DINV2#	+/- 100 mils	DSTBP2#, DSTBN2#	+/- 25 mils
DATA#[63..48], DINV3#	+/- 100 mils	DSTBP3#, DSTBN3#	+/- 25 mils

Source Synchronous ADDRESS :

Address#[63..3], REQ#[4..0], ADSTB#[1..0]			
Transmission Line Type	Total Trace Length	Normal Impedance	Spacing (mils)
Strip-line	1.0 ~ 6.5 inch	55+/-10%	4 & 12

Signals Name	Signals Matching	Strobes associated with the group	Strobe Matching
A#[16..3], REQ#[4..0]	+/- 200 mils	ADSTB0#	+/- 25 mils
A#[31..17]	+/- 200 mils	ADSTB1#	+/- 25 mils

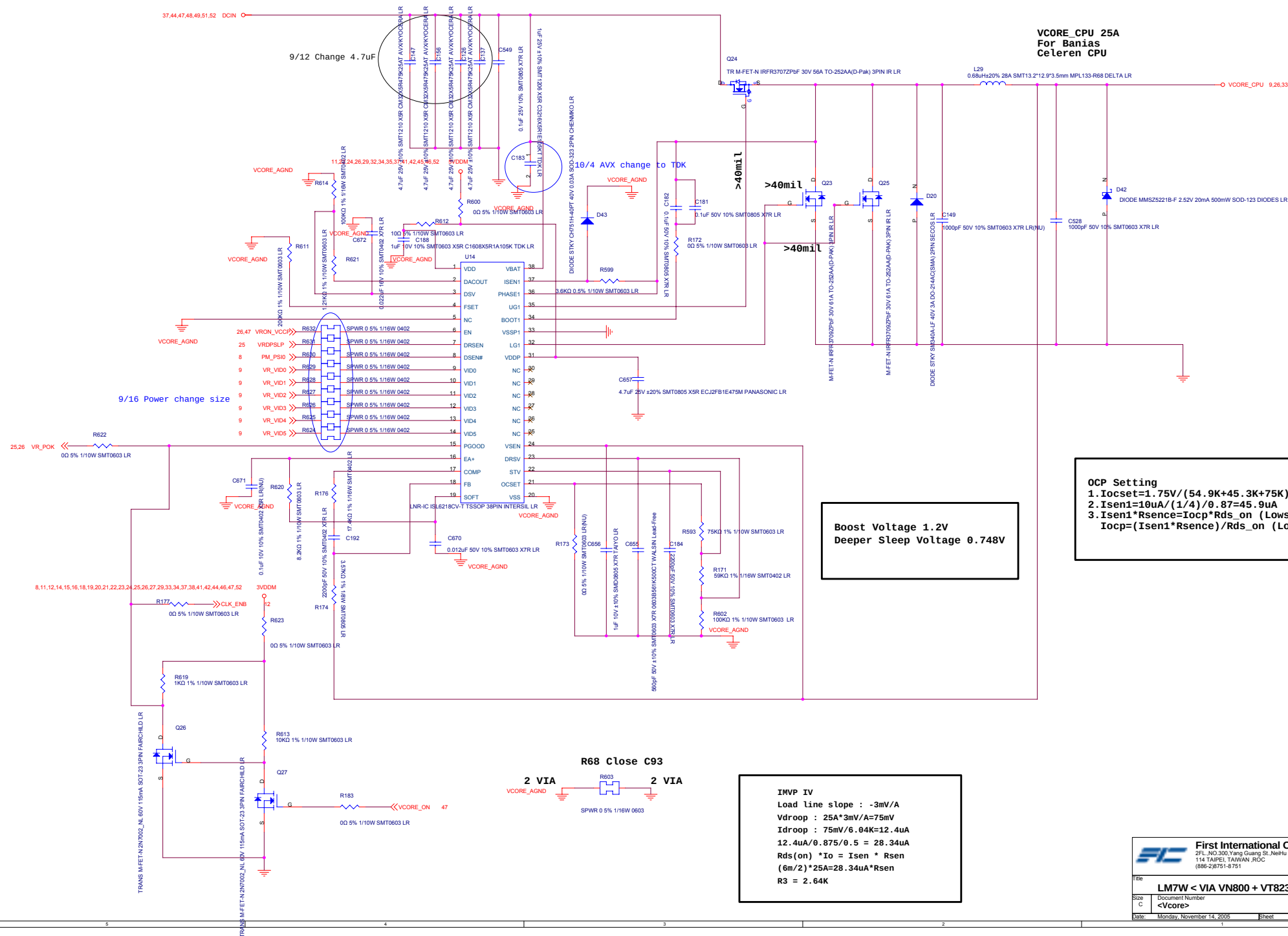


	TEST1	TEST2
Banias processor A0	1K	1K
Banias processor A1 or later	No Stuff	No Stuff

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2FL, NO.300, Yang Guang St., Neihu
114 TAIPEI, TAIWAN, ROC
(886-2)8751-8751

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C	<CPU 1/2>		
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VCORE_CPU



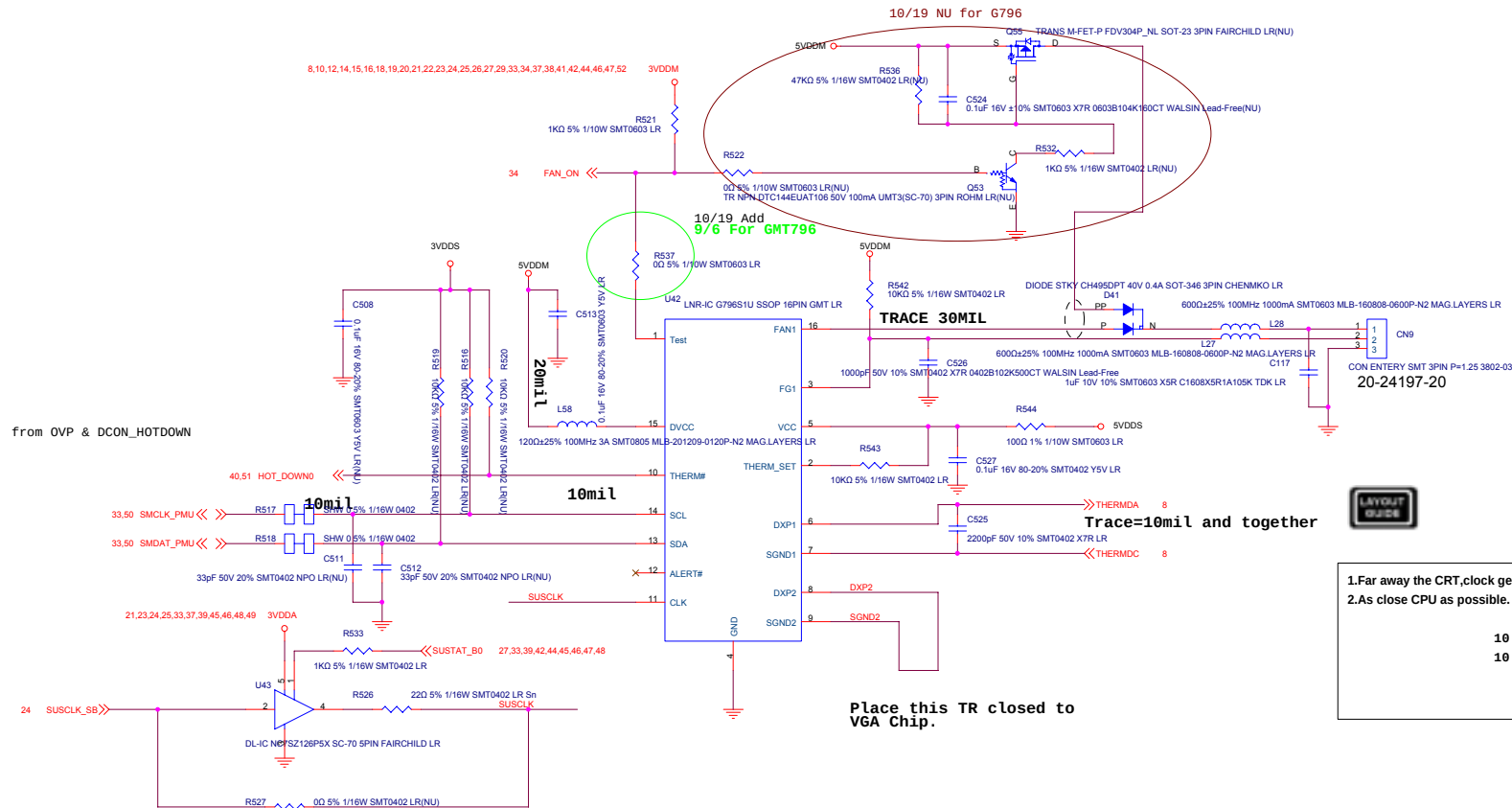
OCp Setting

1. $I_{ocset} = 1.75V / (54.9K + 45.3K + 75K) = 10\mu A$
2. $I_{sen1} = 10\mu A / (1/4) / 0.87 = 45.9\mu A$
3. $I_{sen1} * R_{sense} = I_{ocp} * R_{ds_on} \text{ (Lowside)}$
 $I_{ocp} = (I_{sen1} * R_{sense}) / R_{ds_on} \text{ (Lowside)}$

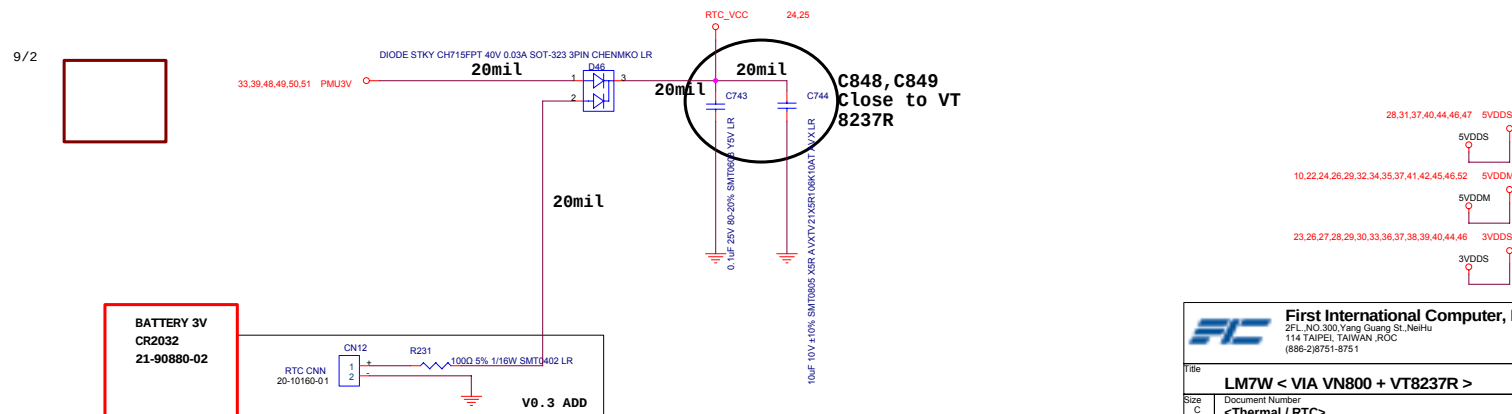
Boost Voltage 1.2V
Deeper Sleep Voltage 0.748V

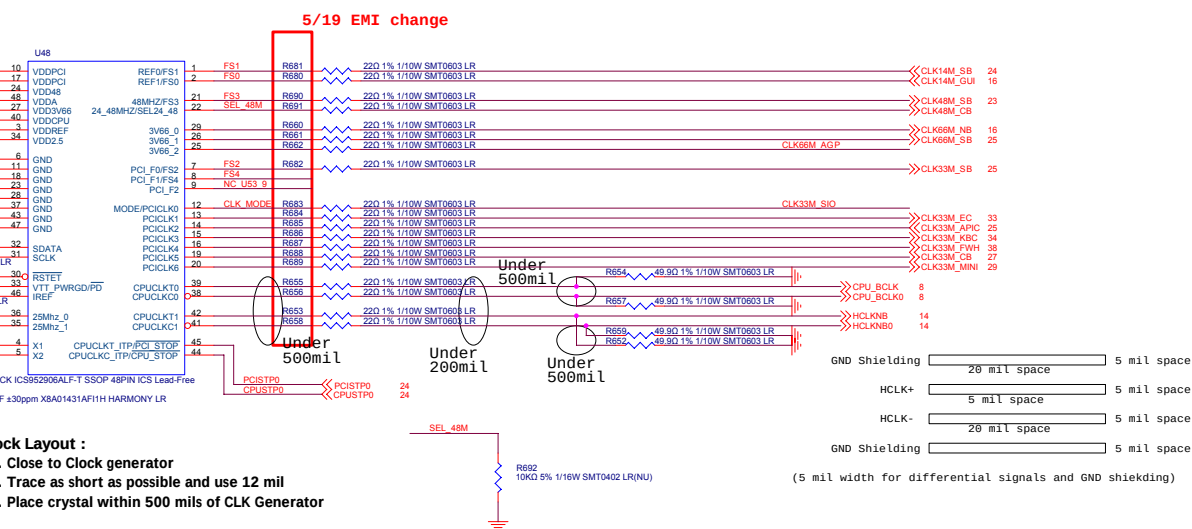
IMVP IV
Load line slope : -3mV/A
Vdroop : $25\text{A} \times 3\text{mV/A} = 75\text{mV}$
Idroop : $75\text{mV} / 6.04\text{K} = 12.4\mu\text{A}$
 $12.4\mu\text{A} / 0.875 / 0.5 = 28.34\mu\text{A}$
 $R_{\text{ds(on)}} \times I_o = I_{\text{sen}} \times R_{\text{sen}}$
 $(6\text{m}\Omega) \times 25\text{A} = 28.34\mu\text{A} \times R_{\text{sen}}$
 $R_3 = 2.64\text{K}$

THERMAL SENSOR

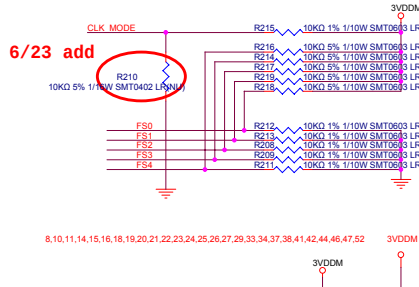
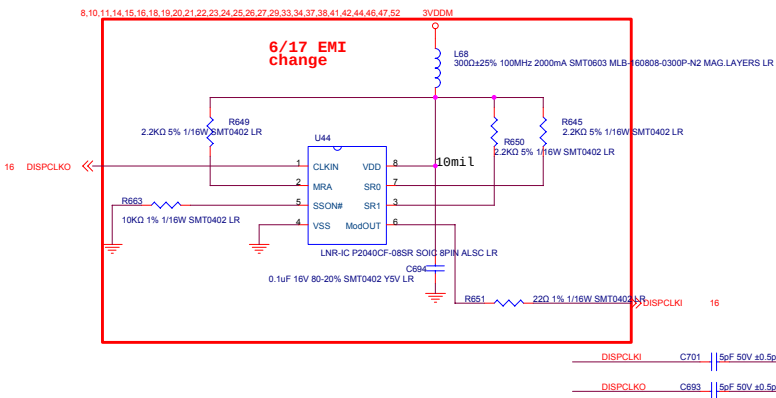
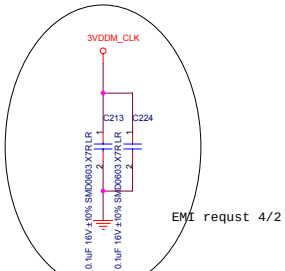


RTC Discharge Circuit

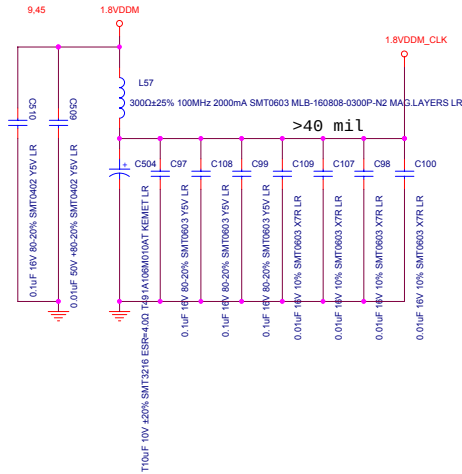
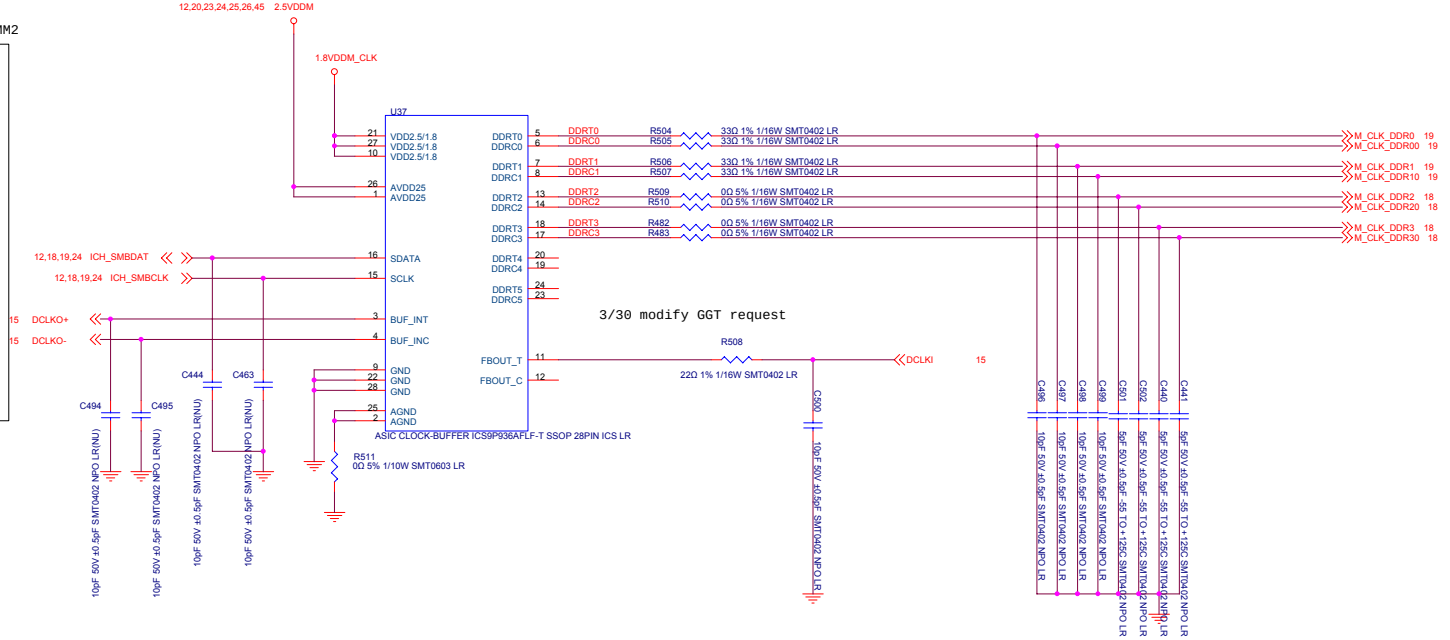
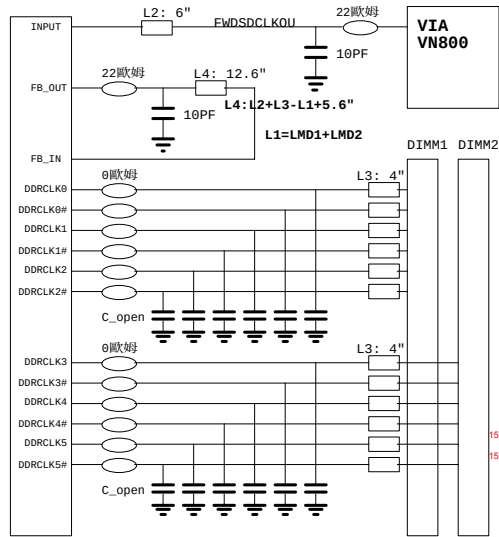


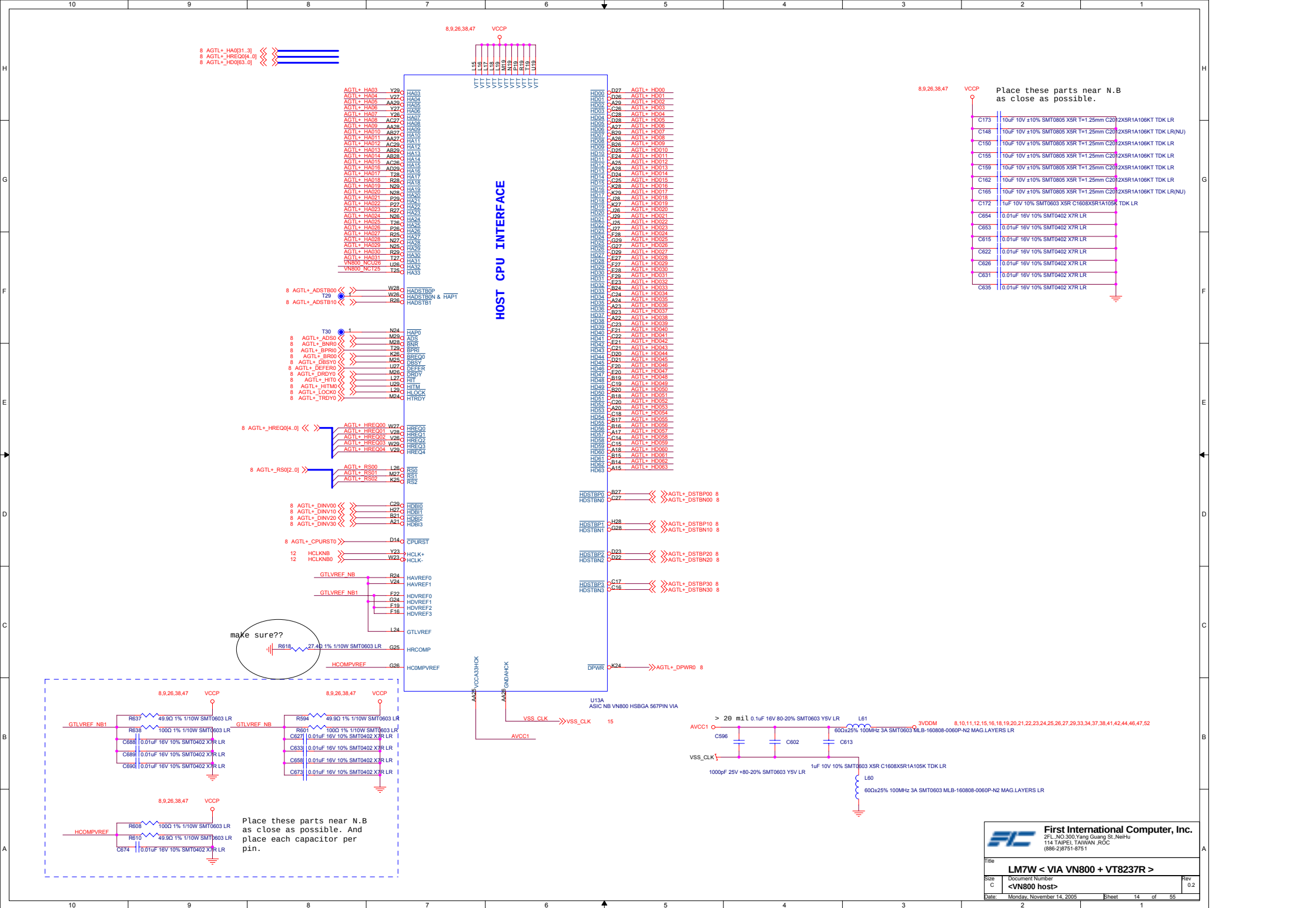


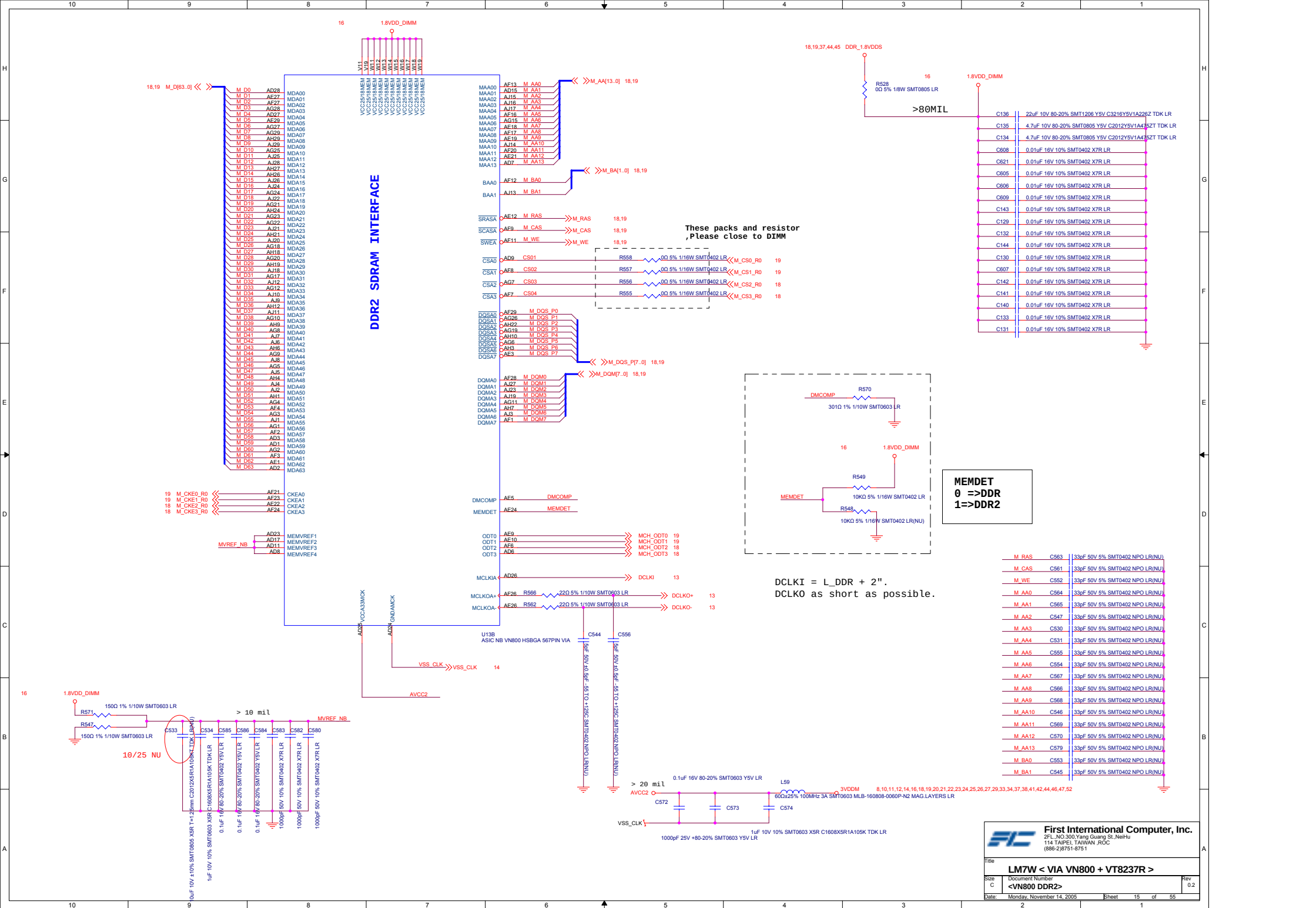
CLOCKS	LENGTH	TRACE / SPACE	NOTES
CPU_BCLK[1..0] MCH_BCLK[1..0] ITP_TV14[1..0]	2" ~ 8"	5 / 20 mils (5 mil space between 1 & 9)	1. Differentials pairs with the same length (within 16 mil) 2. CPU & NB trace mismatch within 20 mil
CLK_ICH66 CLK_MCH66 CLK_AGP	4.5" ~ 9.0" MAX : 8.5"	5 / 20 mils	Length mismatch within 100 mils
CLK_ICHPCI CLK_SIOPCI CLK_FWHPCI	4.5" ~ 9.0"	5 / 20 mils	Length same as CLK66 Clock Length mismatch within 100mils
CLK_MINIPCI CLK_1394PCI CLK_PMU08PCI CLK_CBPCI	4.5"~9.0"	5 / 20 mils	1. Making PCI length with minimum various 2. Length Require CLK33-2.5" 3. Length mismatch +/- 2.0"
CLK_SIO14 CLK_ICH14 CLK_TV14	2.0"~9.0"	5 / 20 mils	1. Length mismatch +/- 500 mils
CLK_ICH48 CLK_MCH48	3.5" ~ 12.5"	5 / 20 mils	

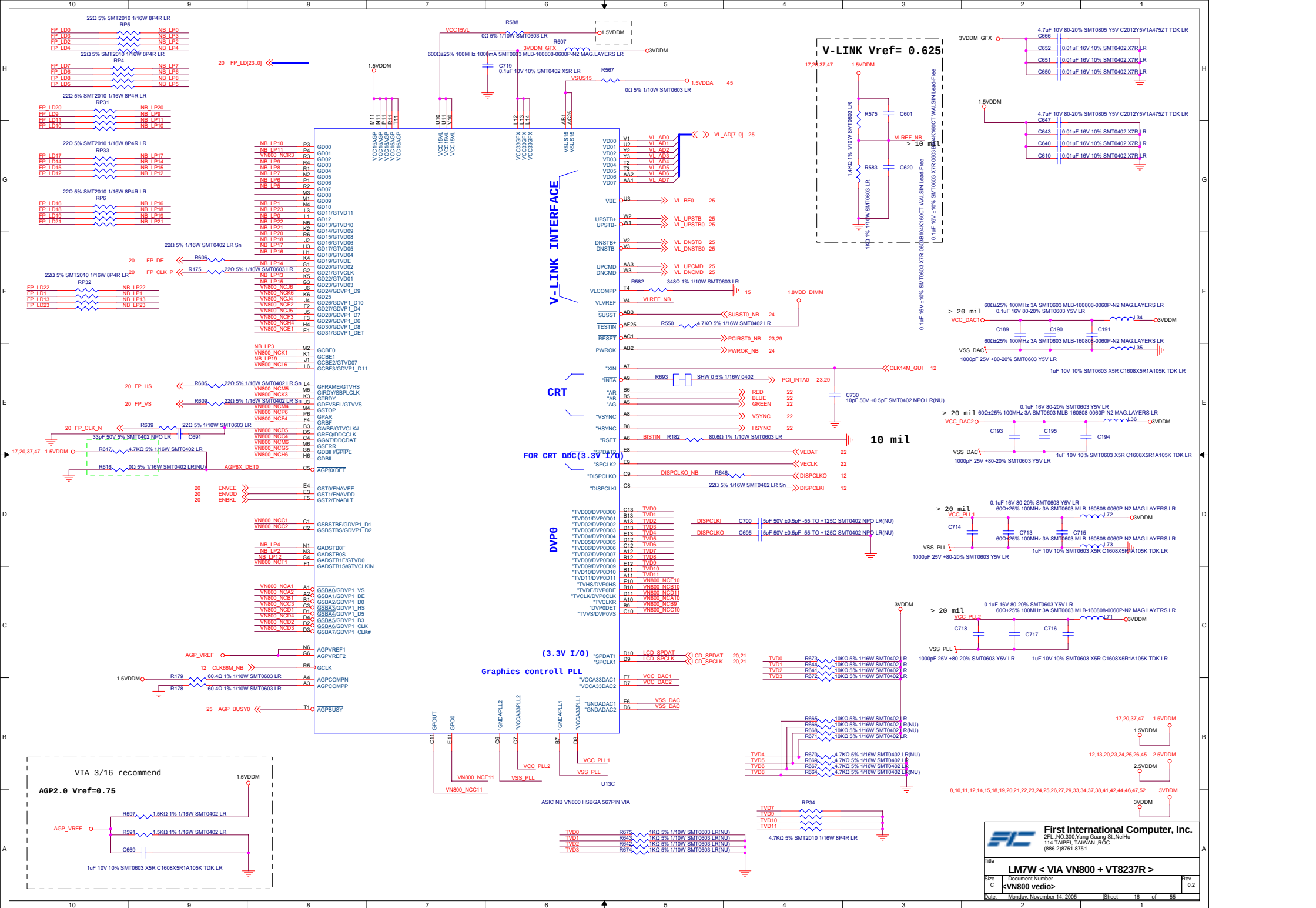
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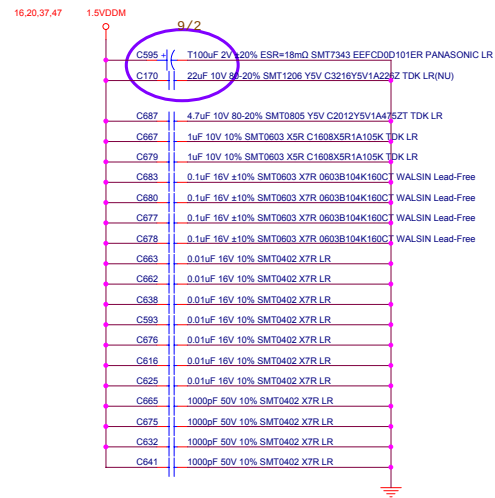
DDR CLOCK BUFFER

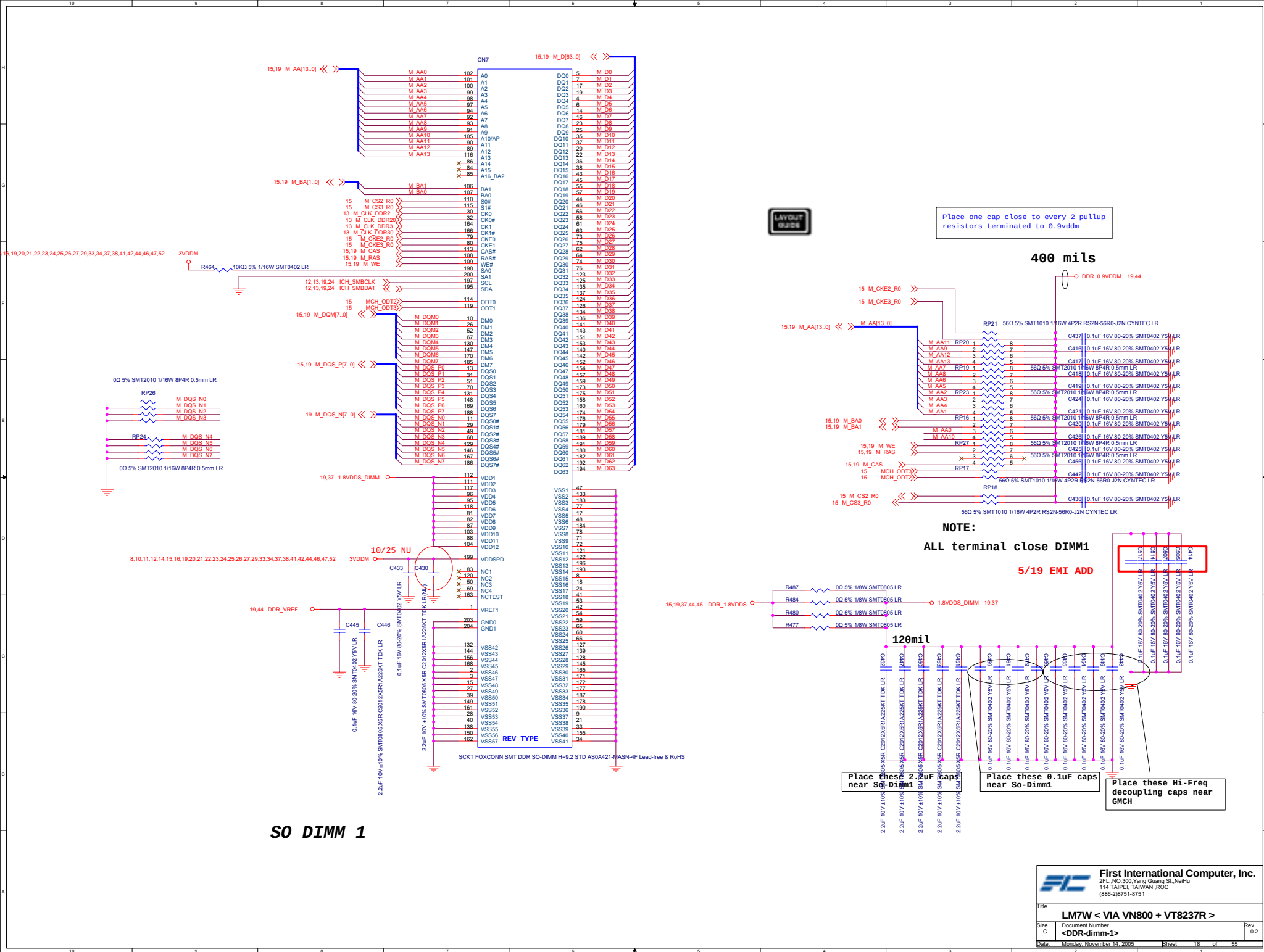


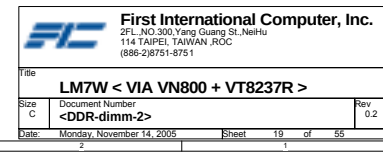


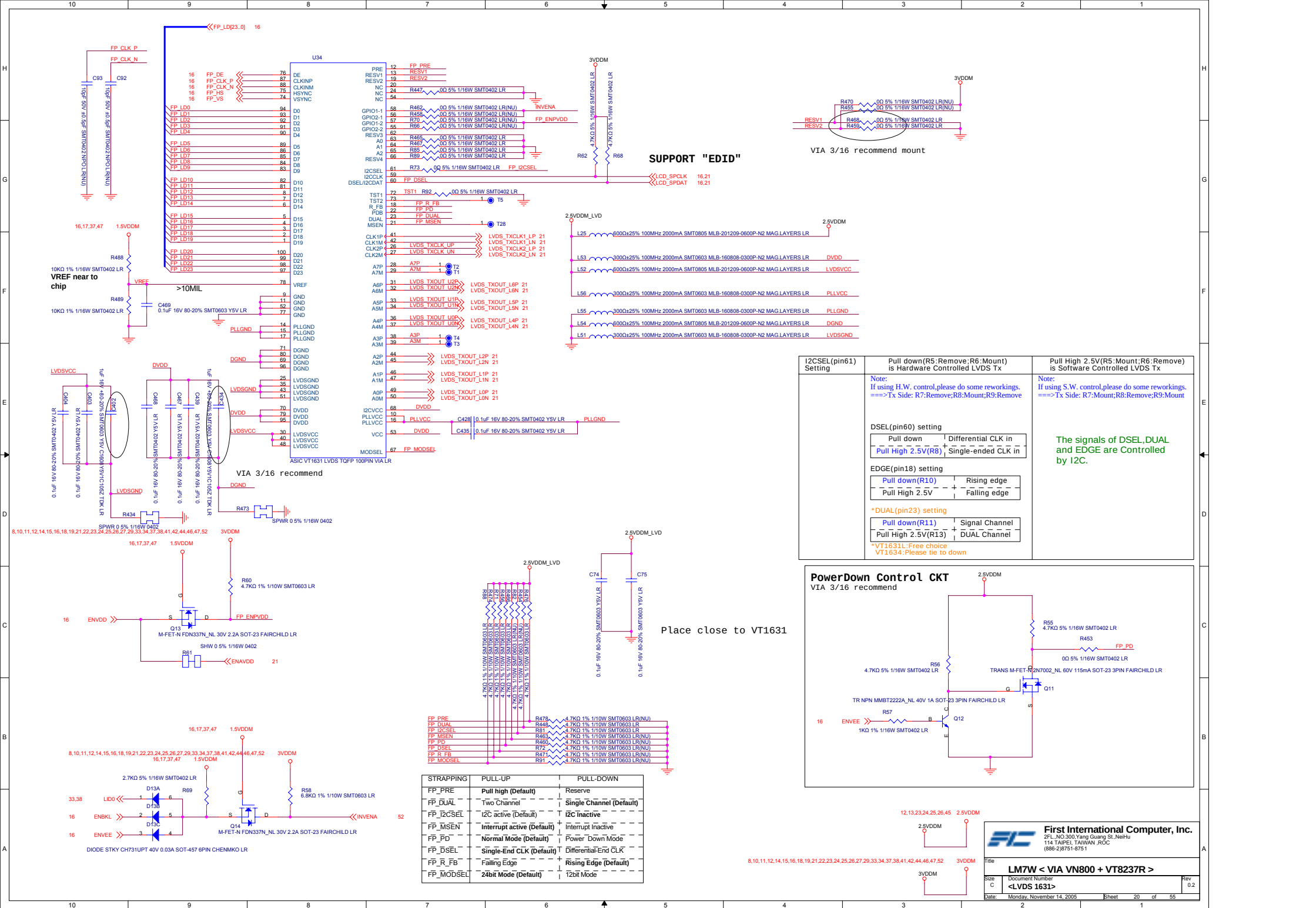


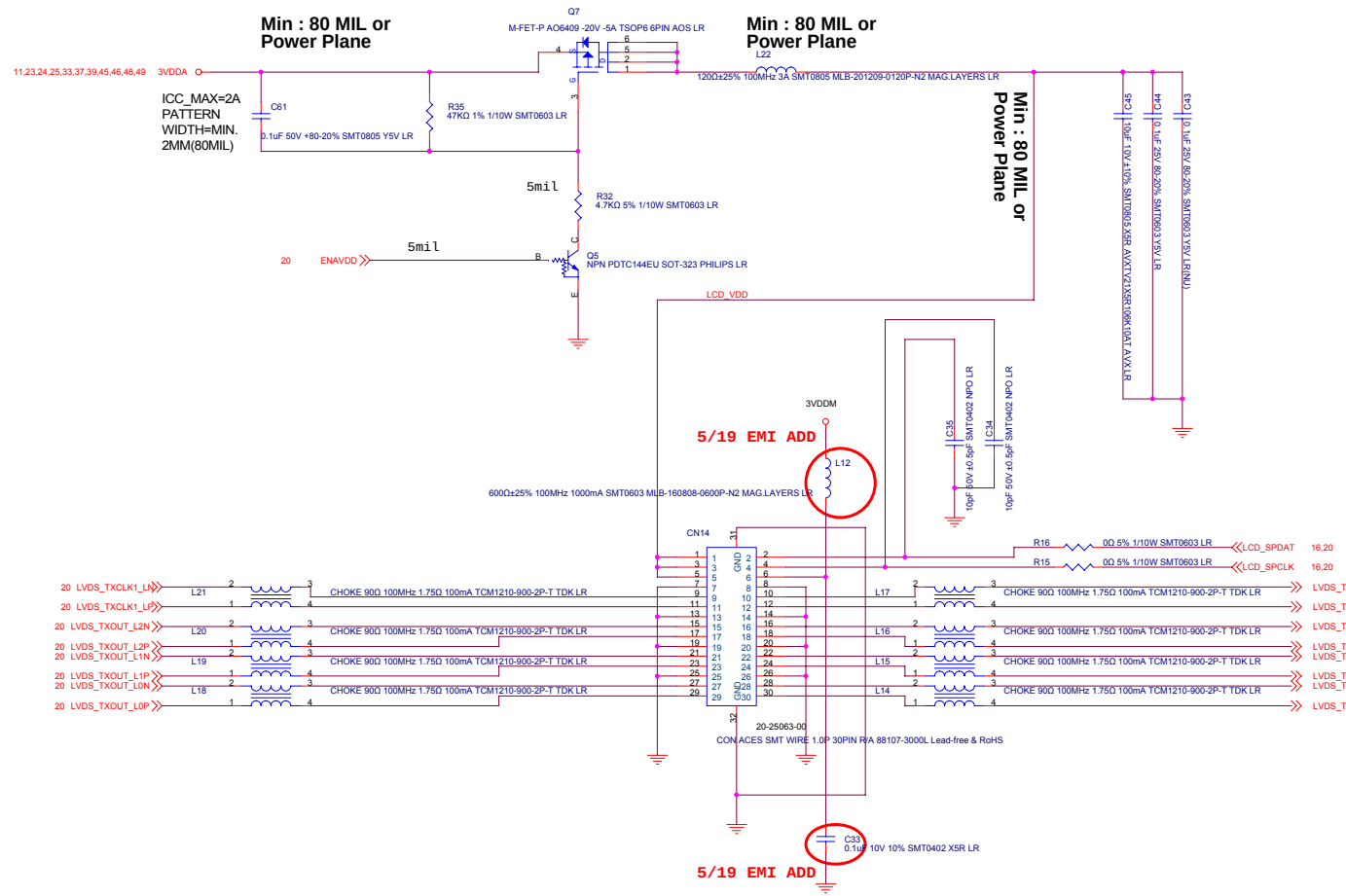






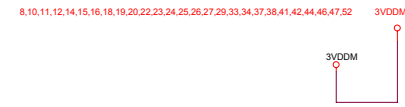






LVDS Interface

Signal	LENGTH	TRACE	SPACE	TRACE MUTCHING	Impedance	Note
LVDS	10"	4 mils (stripline) 6 mils (microstrip)	20 mils (edge to edge) 20 mils (pair to pair) 20 mils (to non LVDS signal)	+/-20 mils (data to clock) +/-20 mils (with a clock pair) X +/-20 mils (clock to clock)	100 ohms +/-15%	Breakout region from NB should be less than 500 mils



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2FL, NO.300, Yang Guang St., NeiHu
114 TAIPEI, TAIWAN, ROC
(886-2)8751-6751

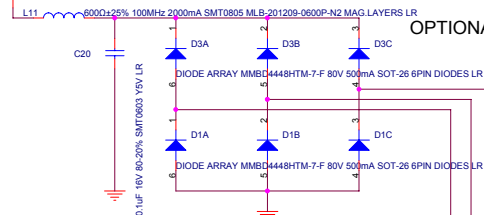
File: LM7W < VIA VN800 + VT8237R >

Size: C Document Number: <LCD connector> Kew: 0.2

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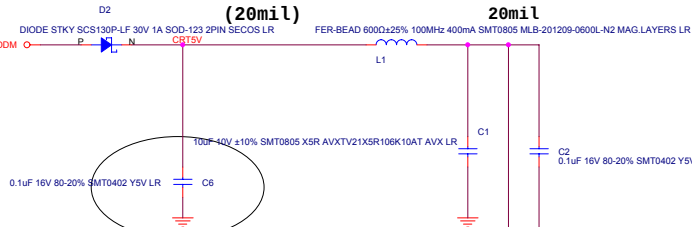
10,11,24,26,29,32,34,35,37,41,42,45,46,52

5VDDM



OPTIONAL ESD PROTECTION DIODES

10,11,24,26,29,32,34,35,37,41,42,45,46,52



16 RED << (10mil)
16 GREEN << (10mil)
16 BLUE << (10mil)



8,10,11,12,14,15,16,18,19,20,21,23,24,25,26,27,29,33,34,37,38,41,42,44,46,47,52

3VDDM

HSYNC >> (10mil)

(10mil)

VSYNC >> (10mil)

(10mil)

DL-IC NC7SZ126PSX_NL SC-70 5PIN FAIRCHILD LR

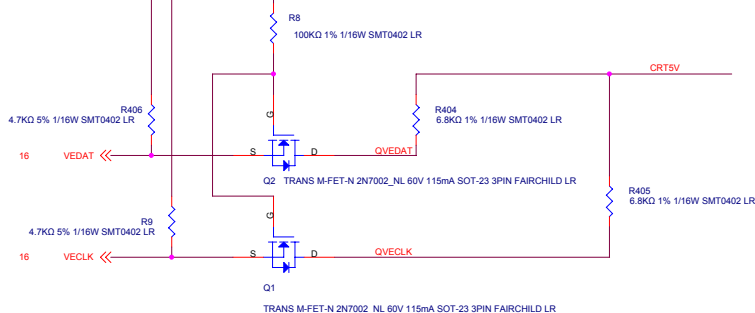
DL-IC NC7SZ126PSX_NL SC-70 5PIN FAIRCHILD LR

R0.4 Modify

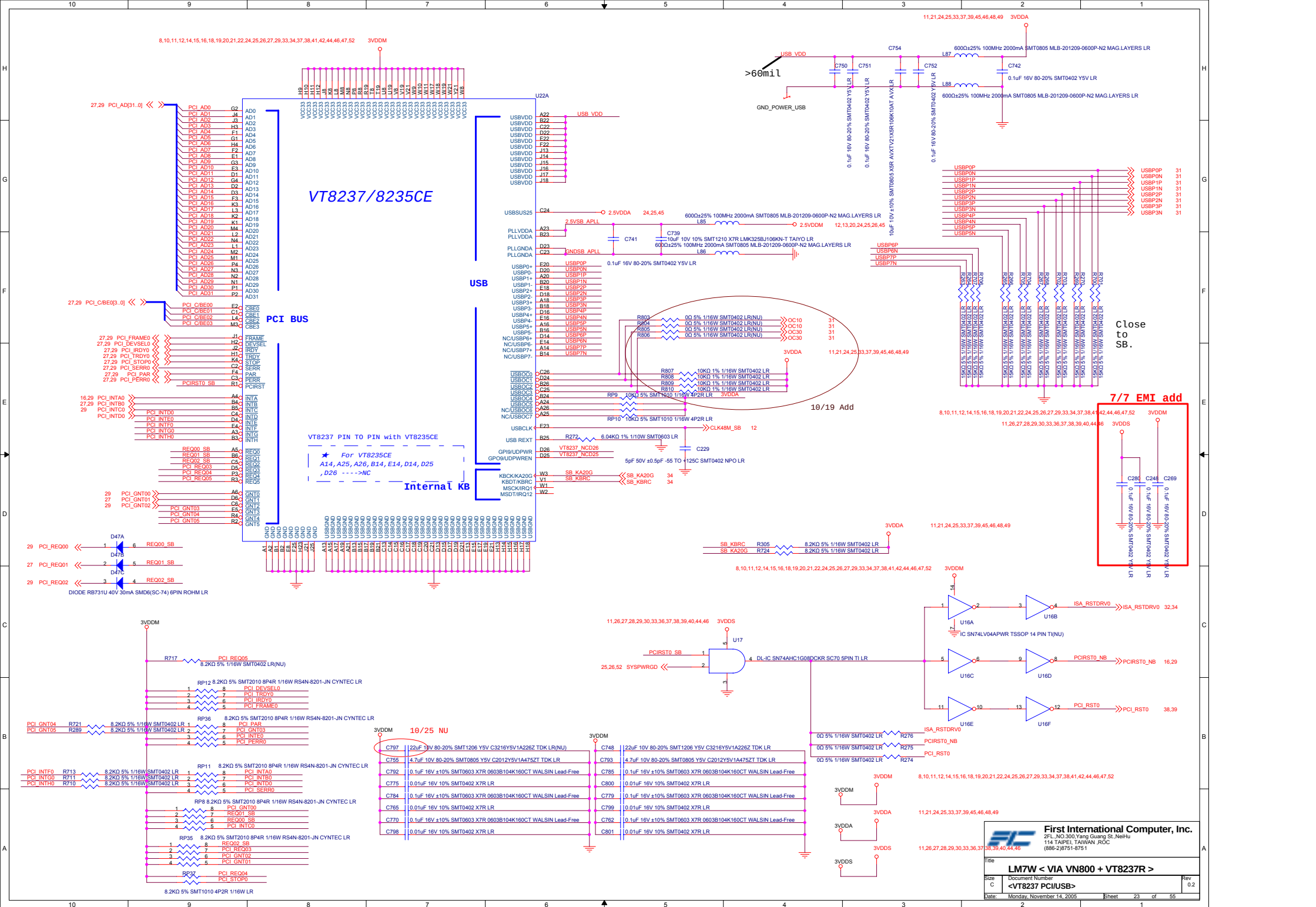
10/20 change to 12pF for VESA SPEC

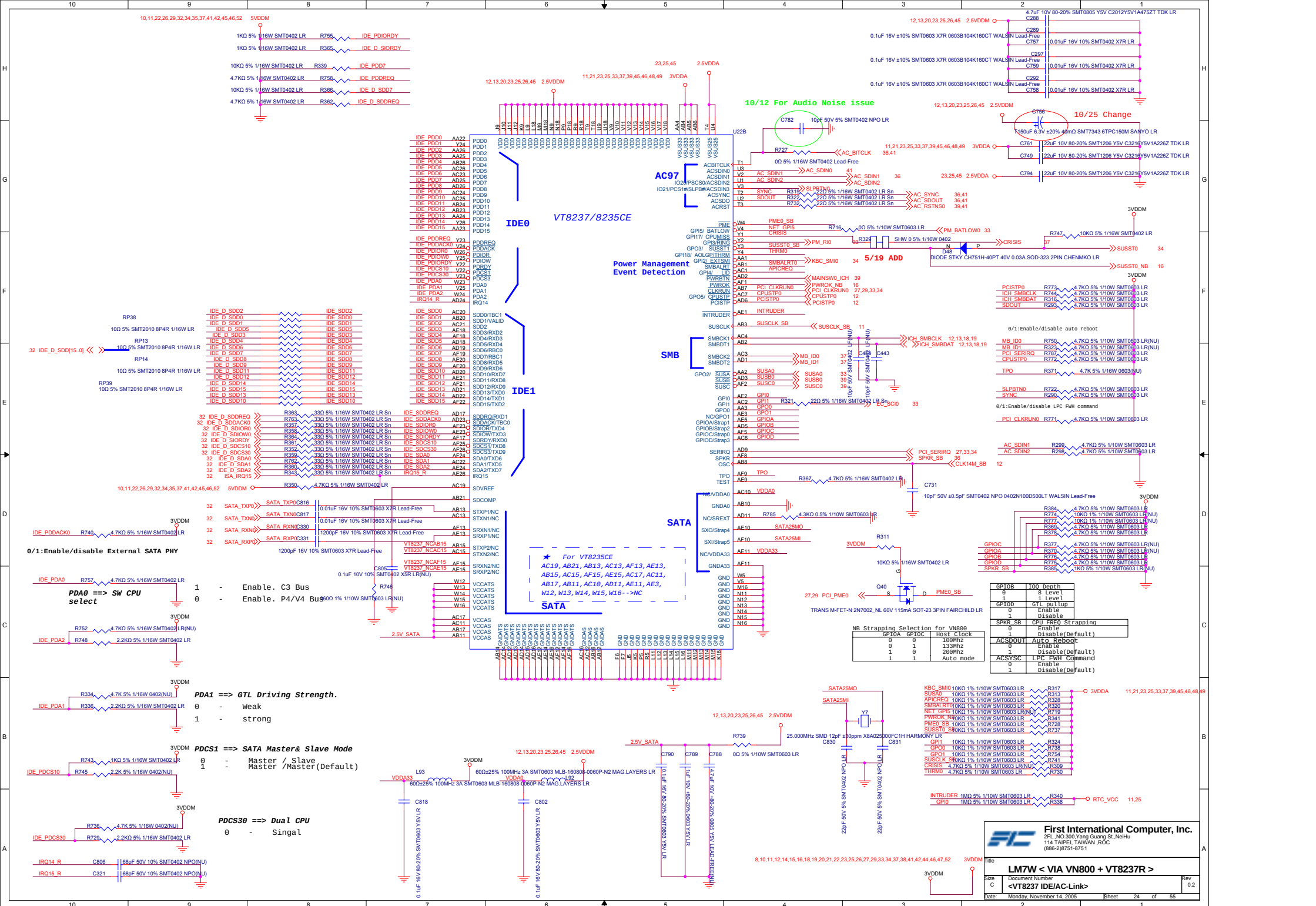
8,10,11,12,14,15,16,18,19,20,21,23,24,25,26,27,29,33,34,37,38,41,42,44,46,47,52

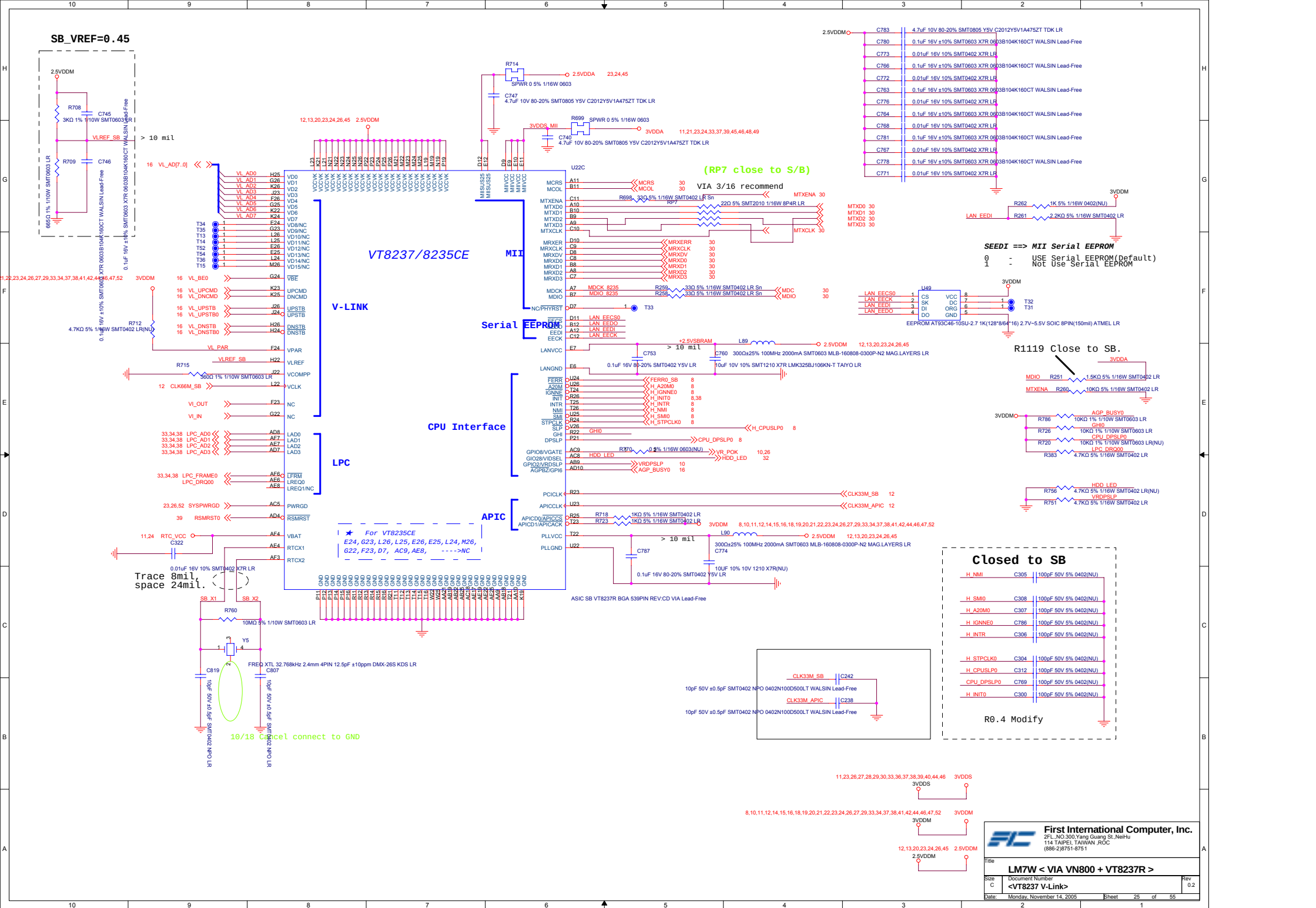
3VDDM



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File		
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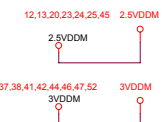


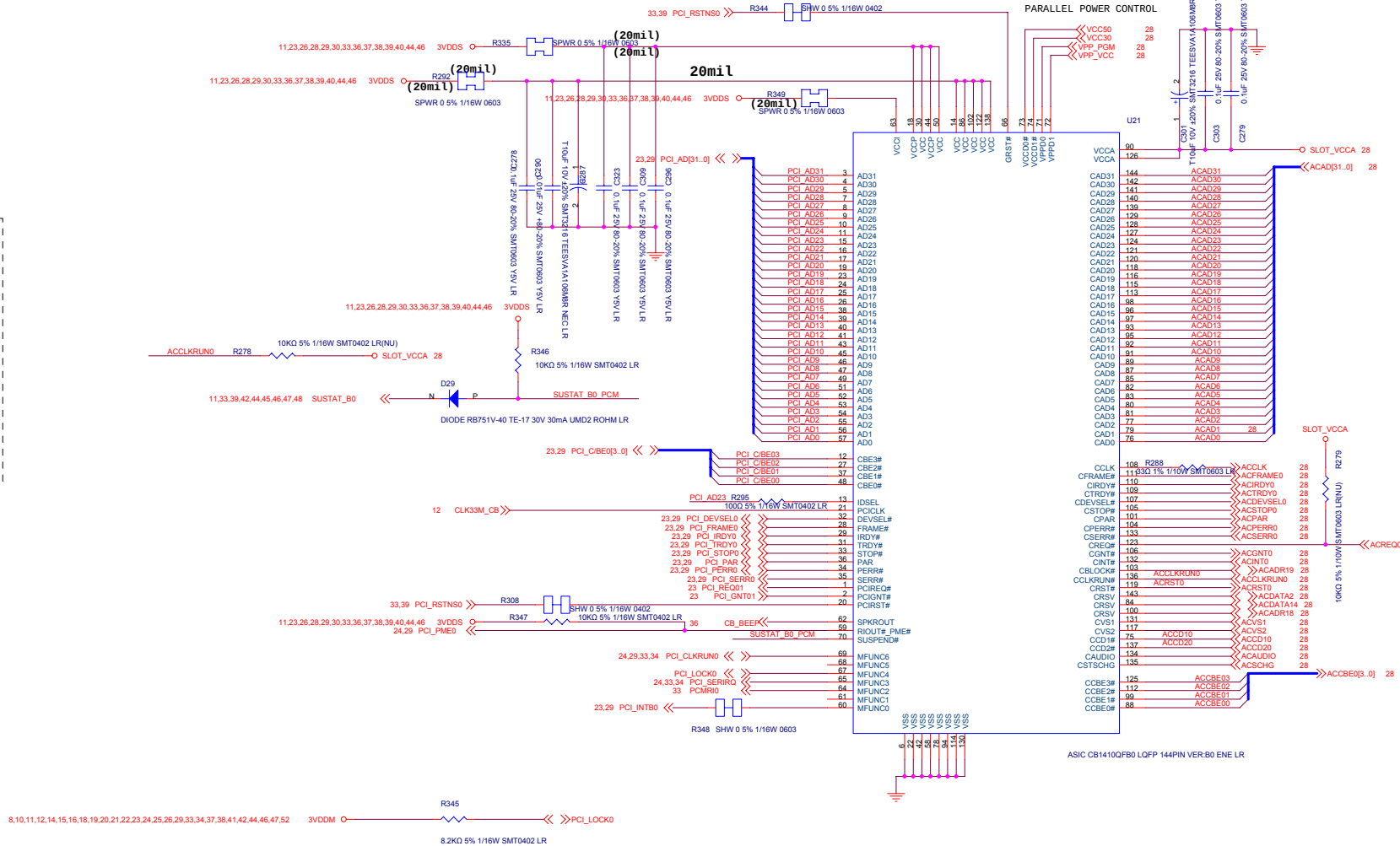


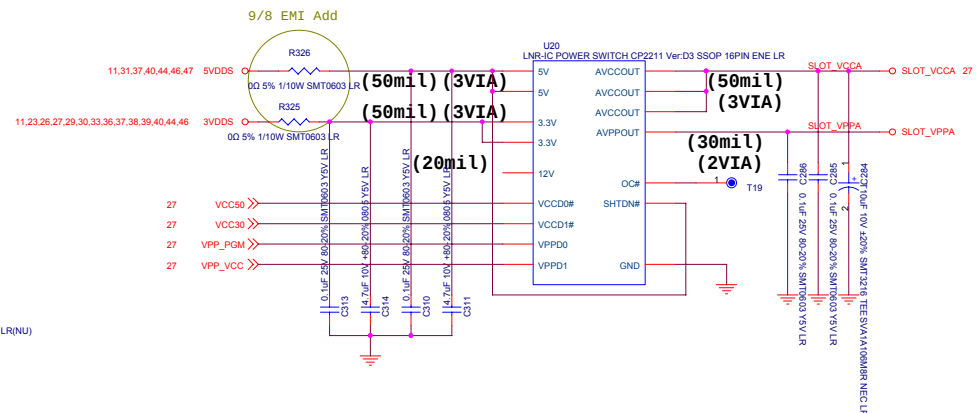
SYSTEM POWER OK CIRCUIT



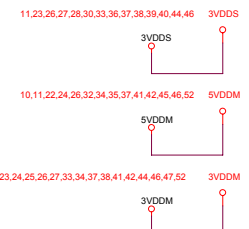
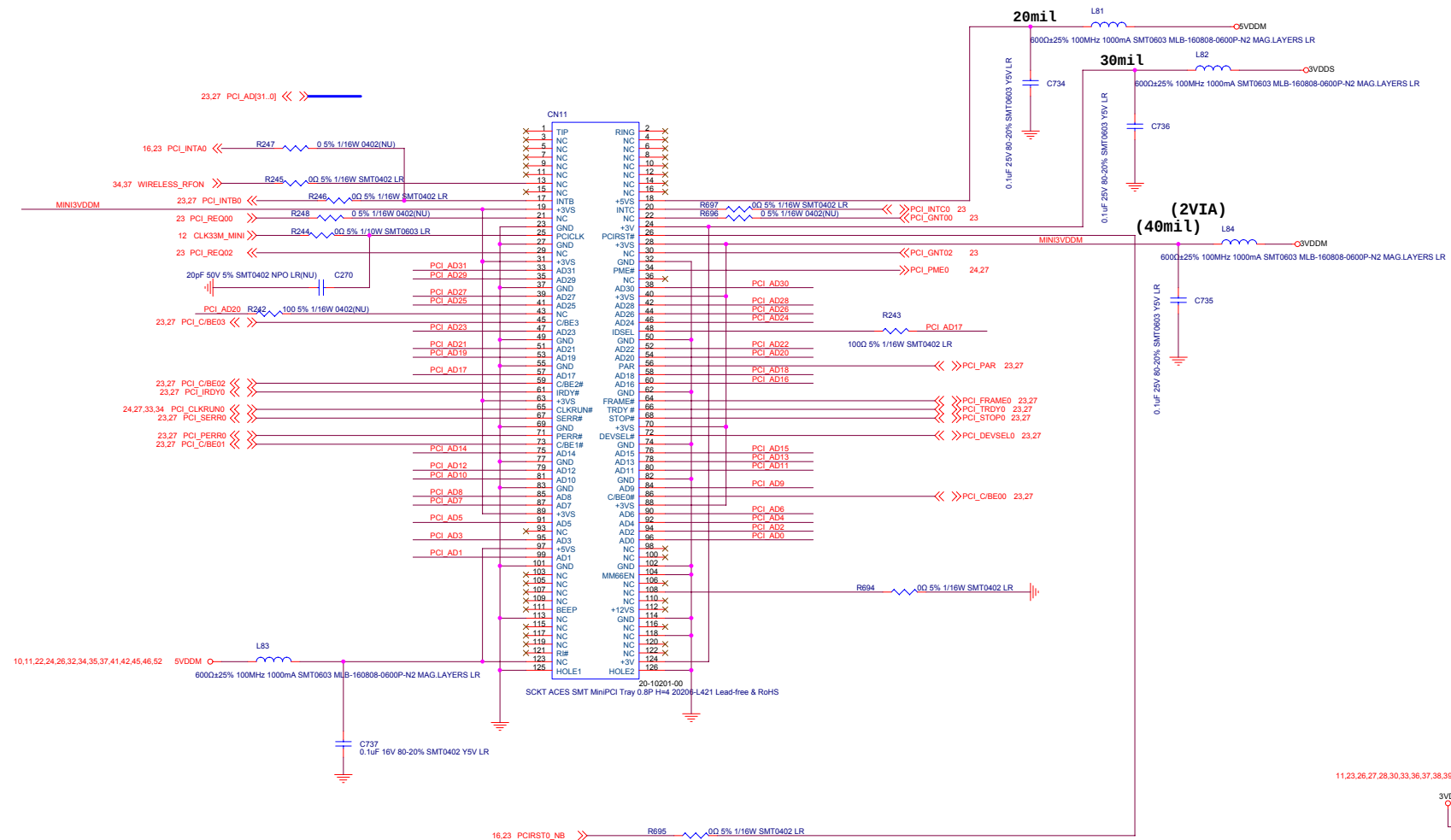
SYSTEM POWER OK CIRCUIT

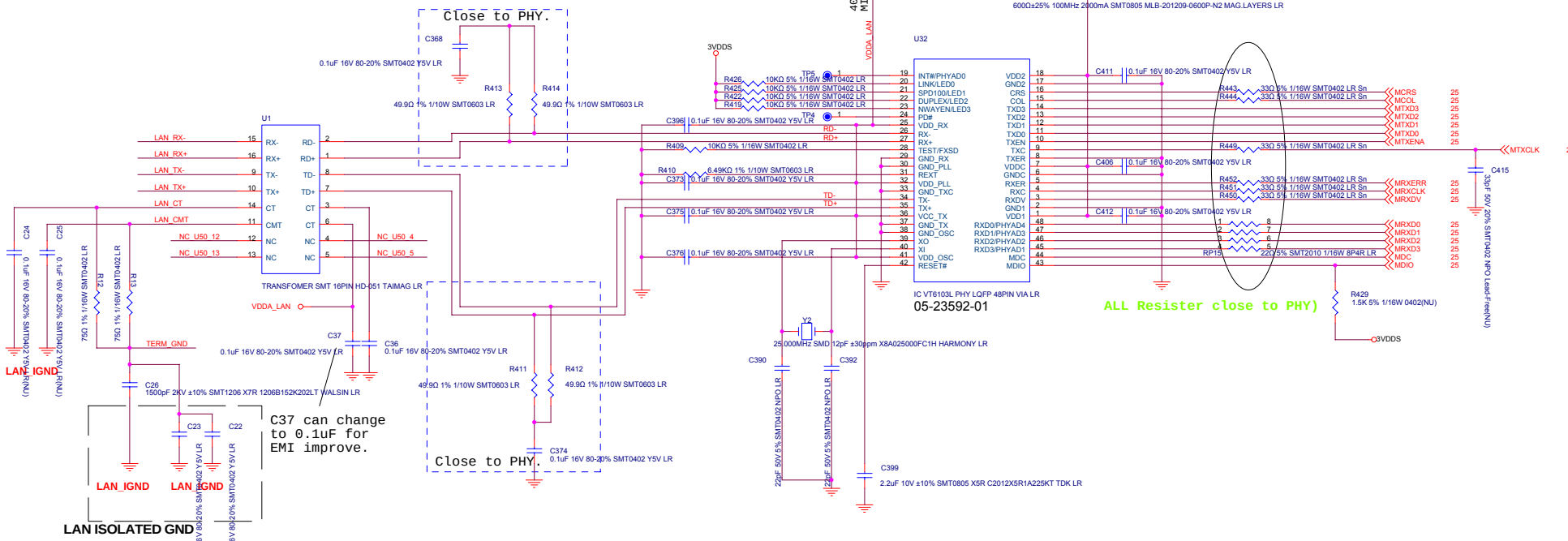






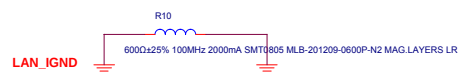
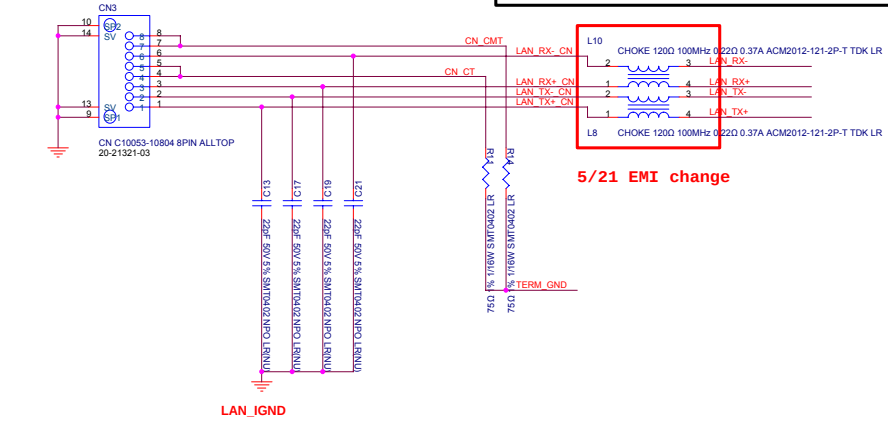
TYPE III MODEM / LAN CONNECTOR





LAN RJ45 JACK

TX & RX layout guide		
other	24 MIL	
TX+	10 MIL	6 MIL
TX-	24 MIL	6 MIL
other		





6mils

6mils

6mils

6mils

TX

RX

* Zdif = 100 Ohm +/- 10%. TX & RX should refer GND.No via & stubs.The Best layer is Top.

* TX/RX trace length < 2 inchs.

* TX+/- need matching trace ± 10 mils length.

* RX+/- need matching trace +10 mils length.

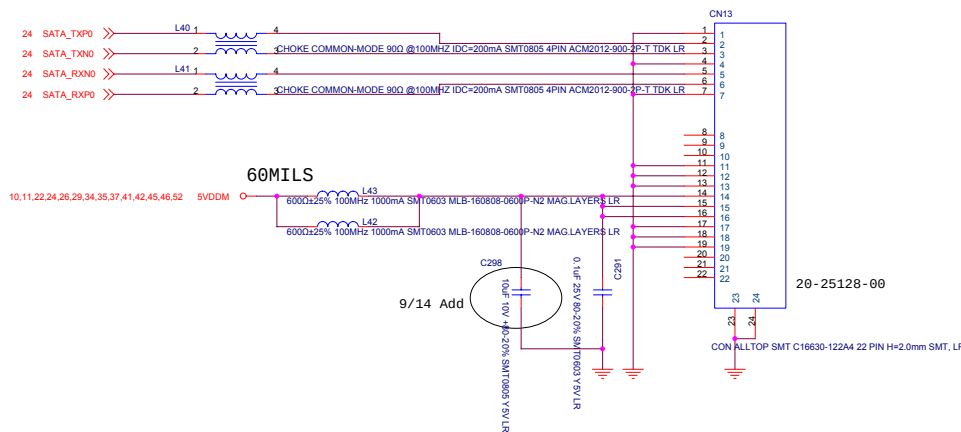
* SATA Pair to Pair Trace matching trace ± 10 mils length.

NOTE

```
SATA differential stripline 20:5:6:5:20
```

SATA differential microstripline 20:6:6:6:20

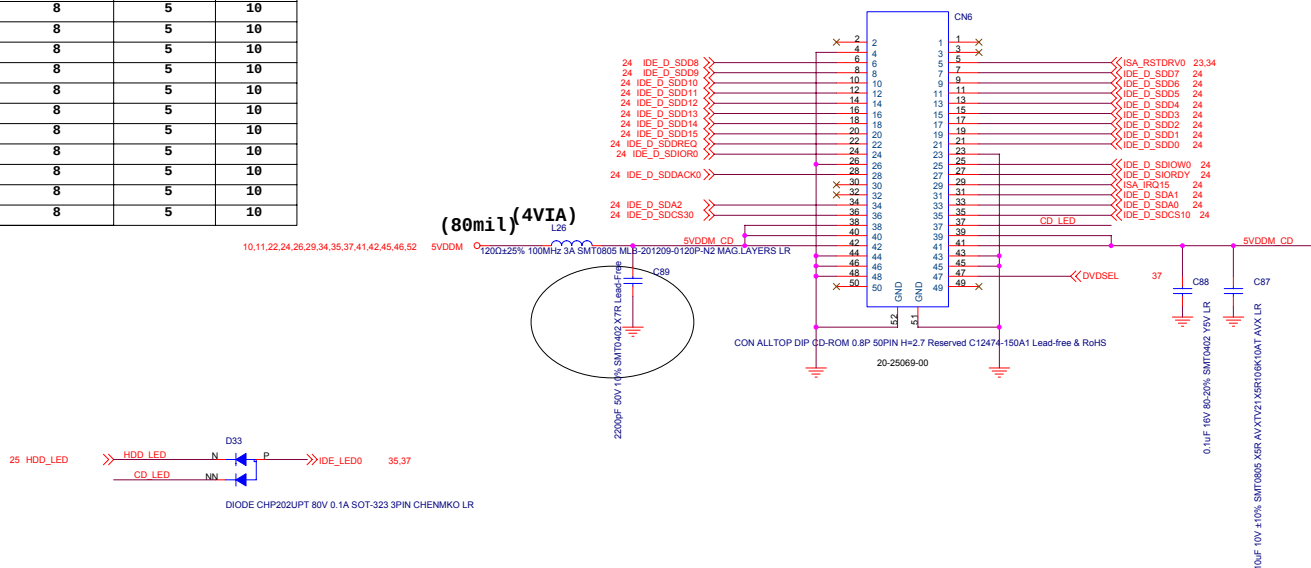
請包GROUND



IDE Signals

Signals	MAX Length (inch)	Width (mils)	Space (mils)
IDE_PDD[15:0]	8	5	10
IDE_SDD[15:0]	8	5	10
IDE_PDA0-2	8	5	10
IDE_SDA0-2	8	5	10
IDE_PDCS 10-30#	8	5	10
IDE_PDDREQ	8	5	10
IDE_SDDREQ	8	5	10
IDE_PDIOW#	8	5	10
IDE_PATADET	8	5	10
IDE_SATADET	8	5	10
IDE_PDDACK#	8	5	10
IDE_SDDACK#	8	5	10

CDROM CNN



Title

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PMU08

CLOSE TO PIN 1 OF PMU08

ATTENTION

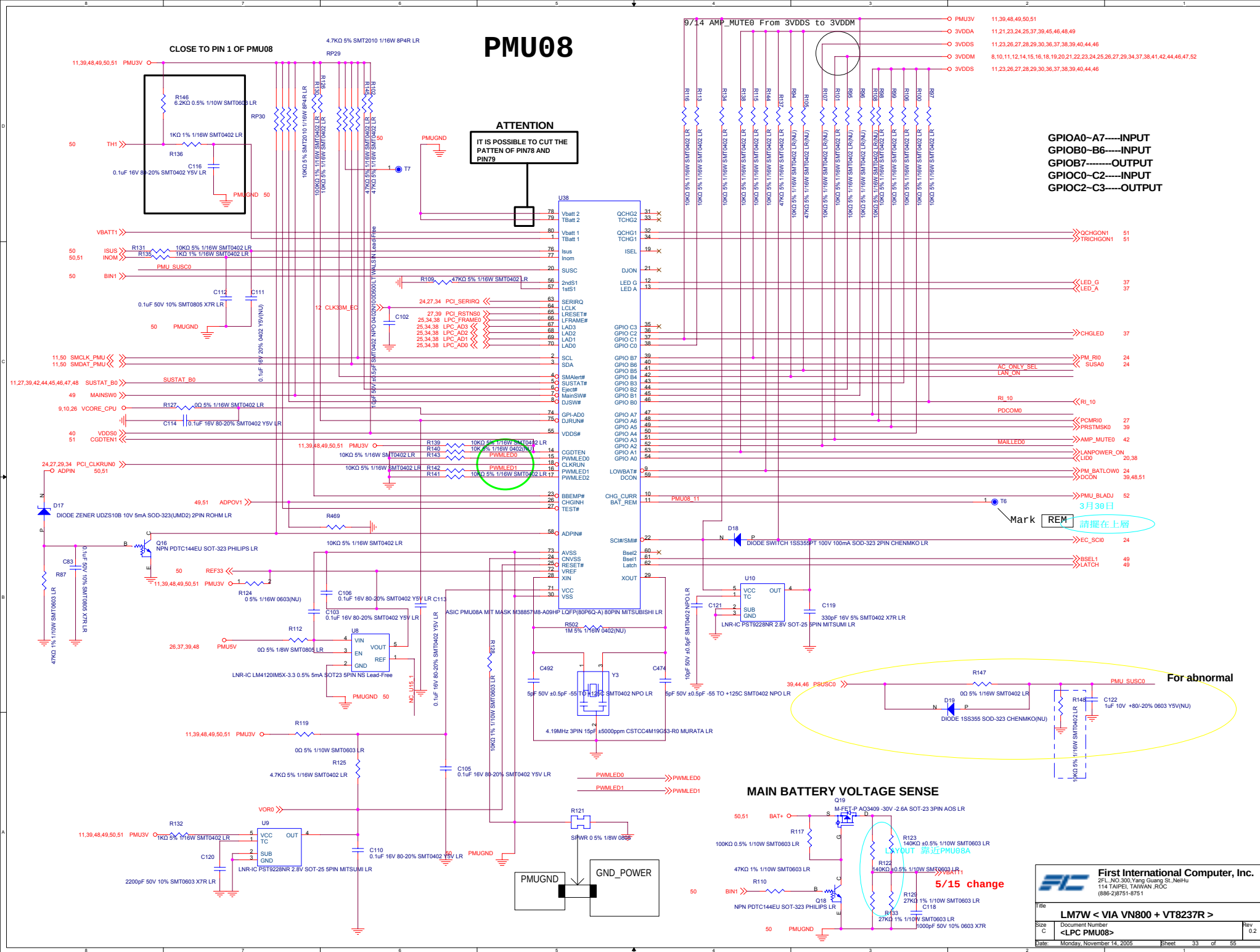
IT IS POSSIBLE TO CUT THE PATTERN OF PIN78 AND PIN79

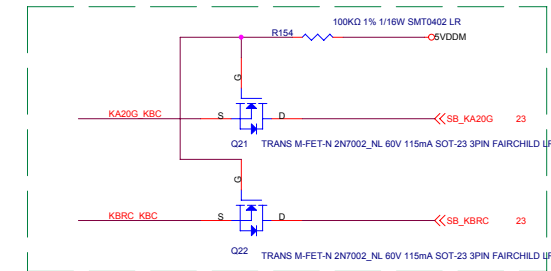
GPIOA0~A7-----INPUT
GPIOB0~B6-----INPUT
GPIOB7-----OUTPUT
GPIOC0~C2-----INPUT
GPIOC2~C3-----OUTPUT

MAIN BATTERY VOLTAGE SENSE

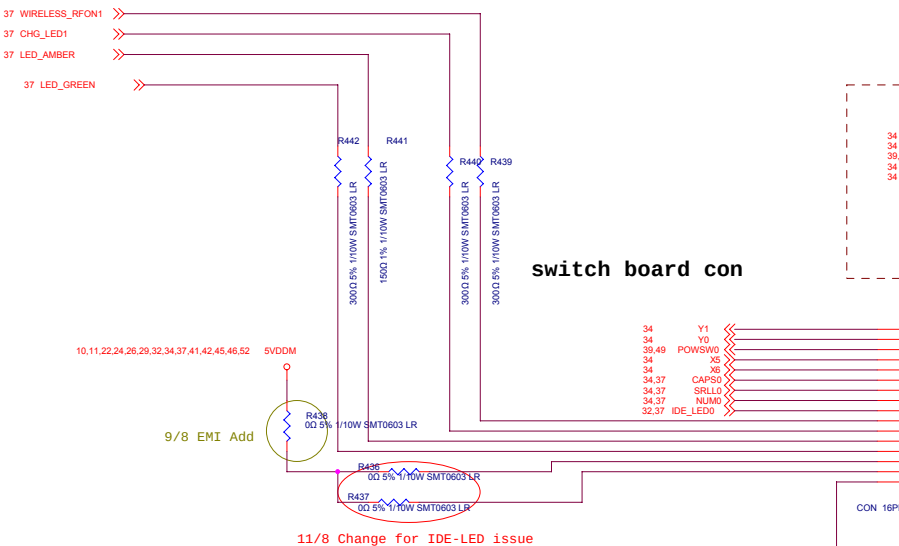
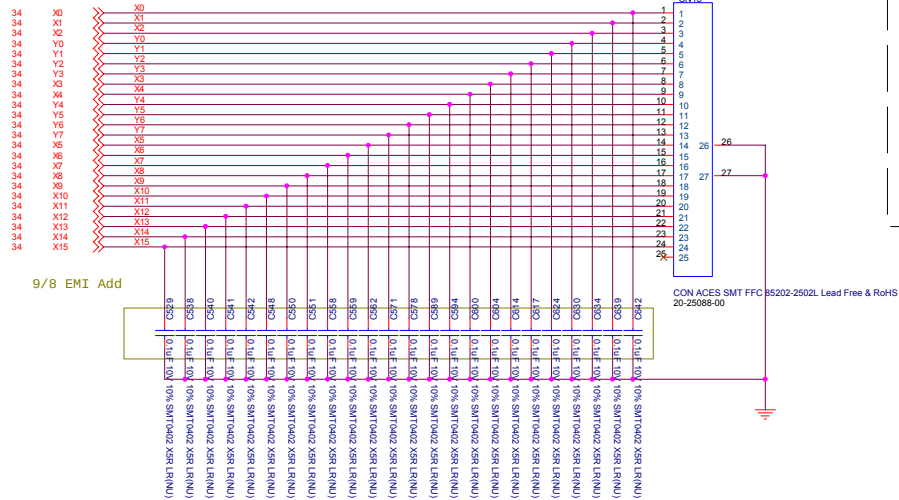
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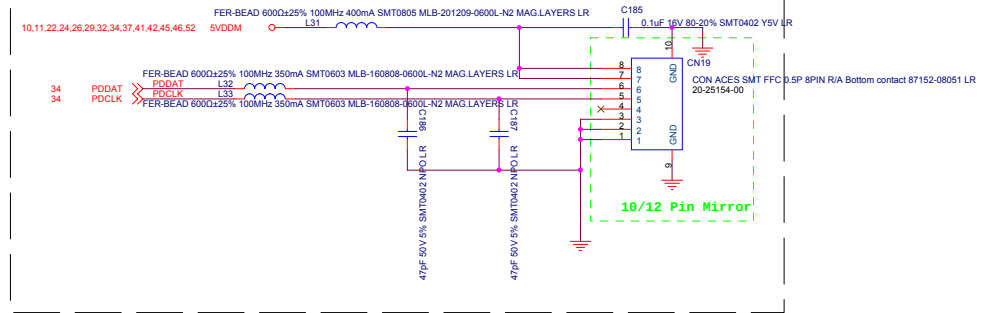


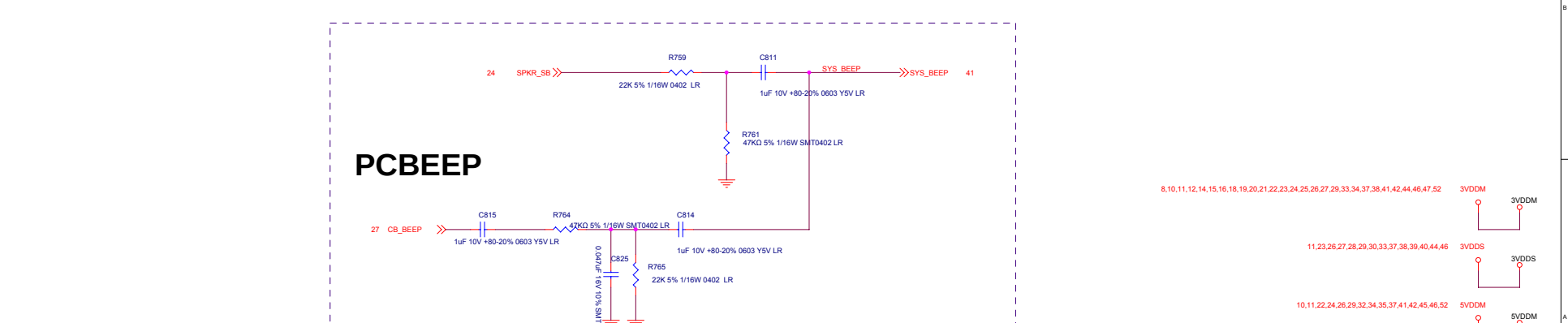
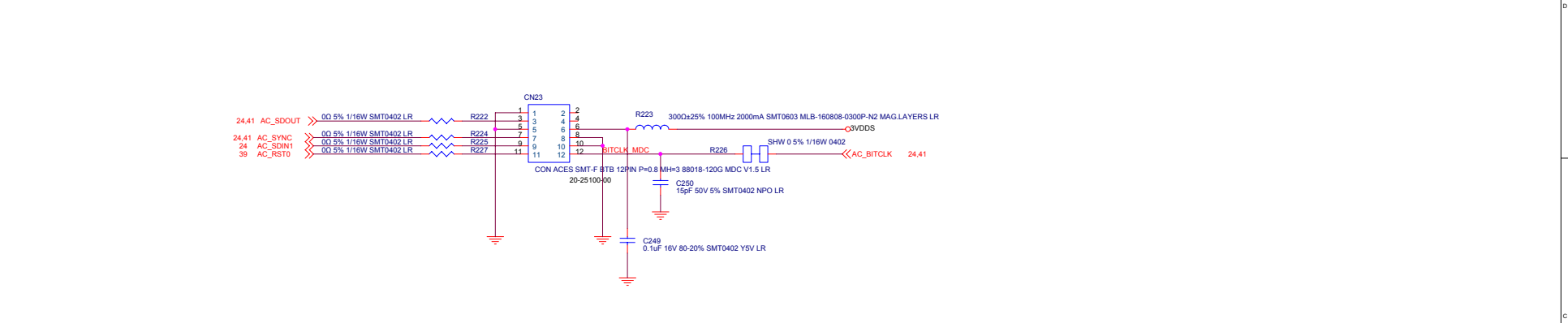


INT KB CNN

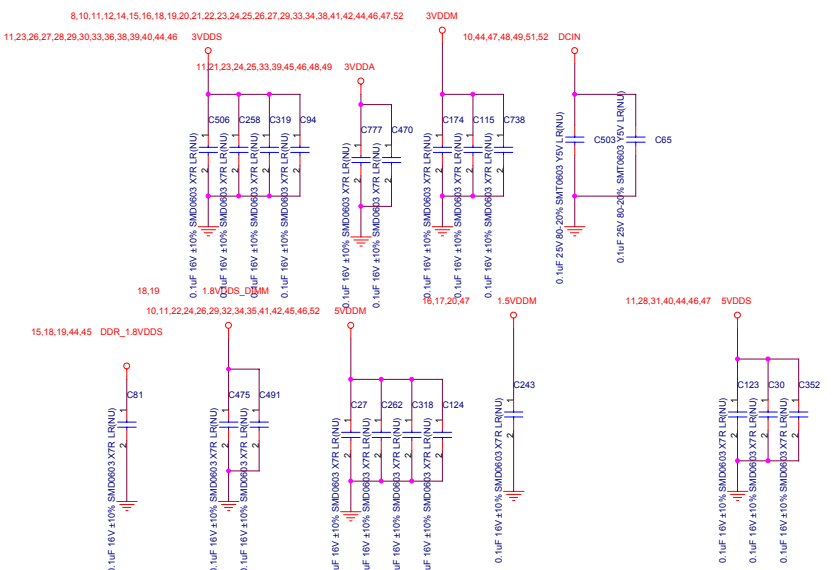
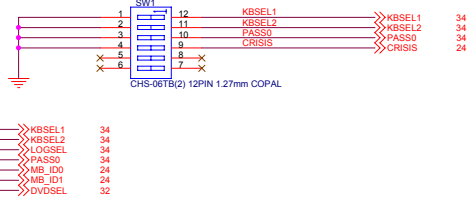
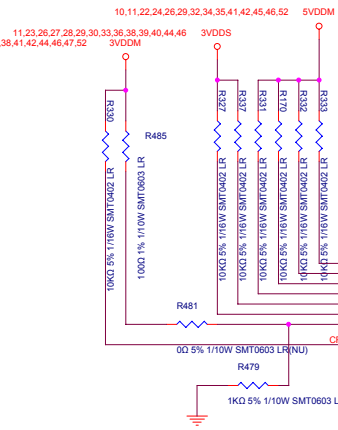


GLIDE PAD CONNECTOR

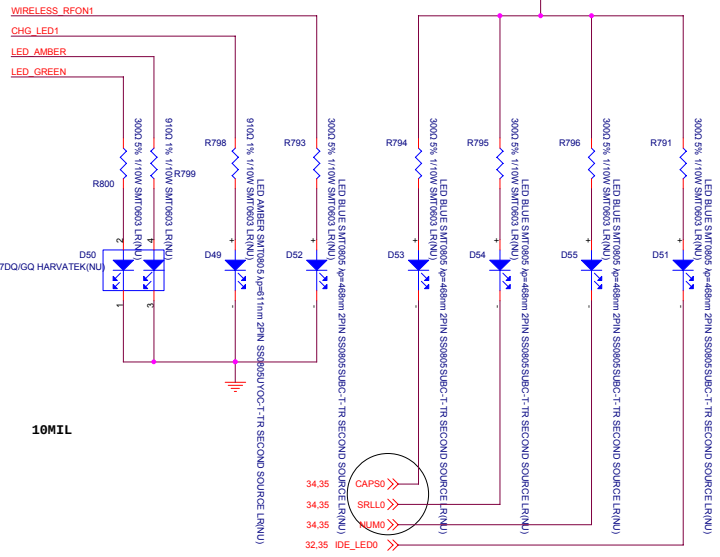
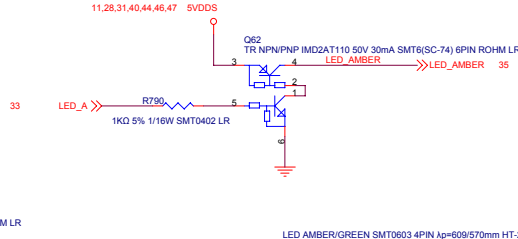
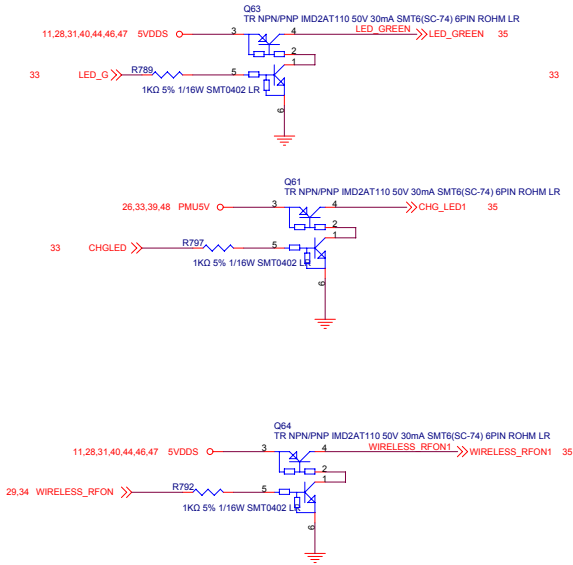




DIP SWITCH



LED indicator control logic

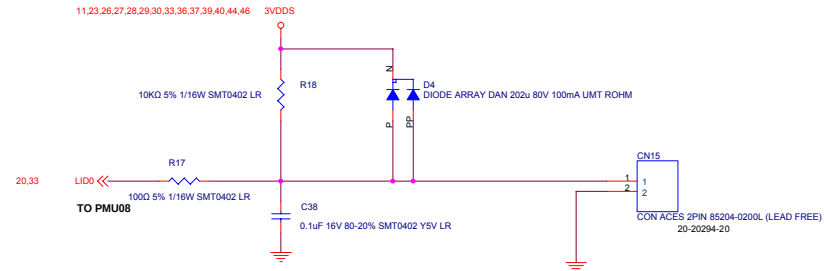


from KBC

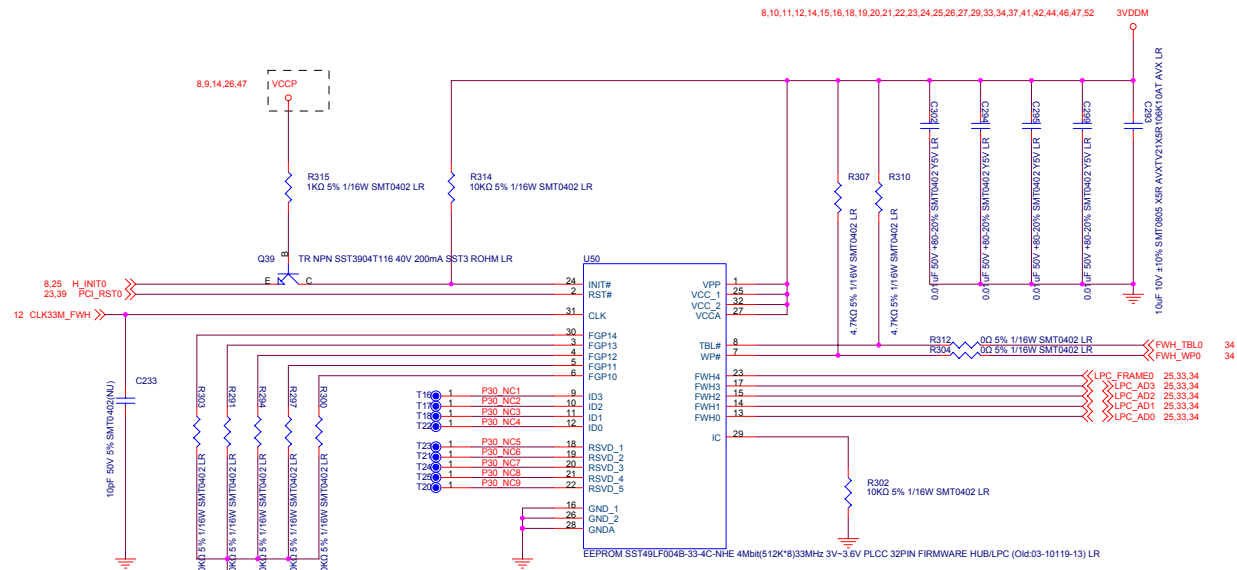
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LID Switch

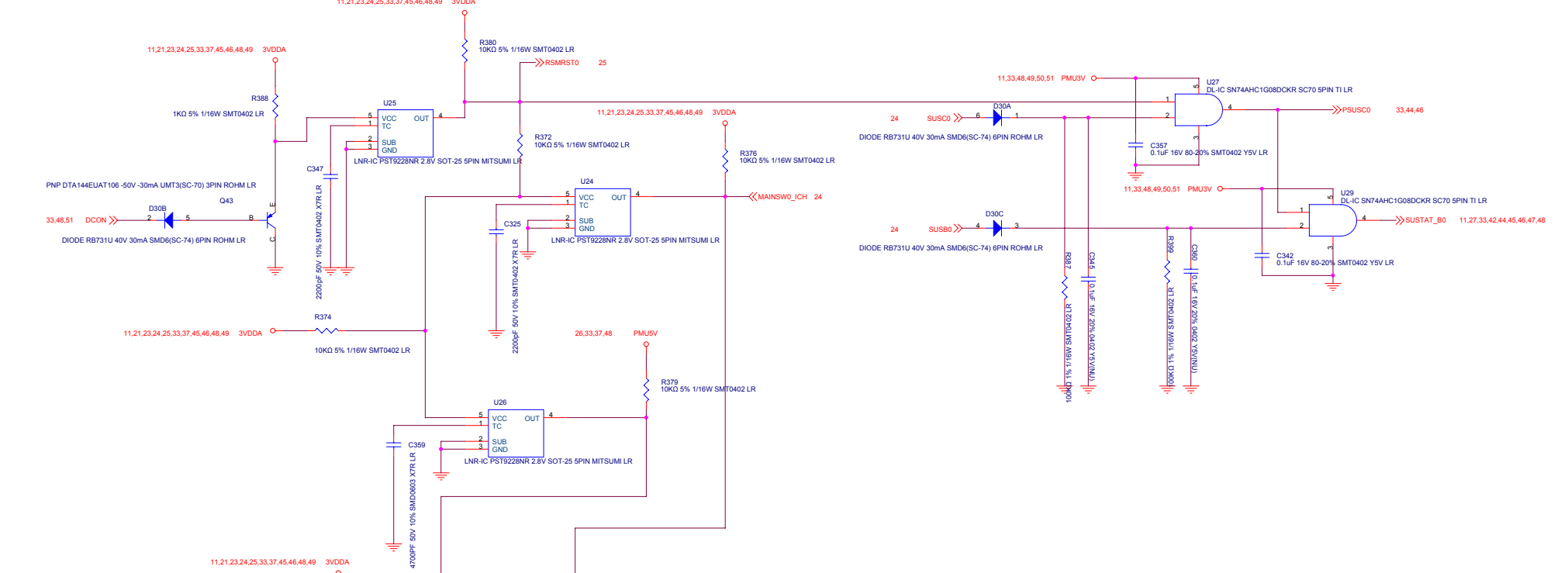


4M FLASH ROM

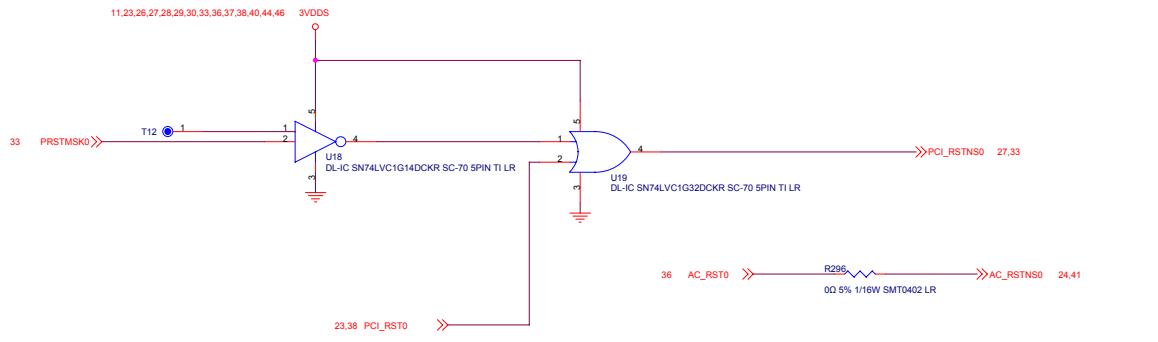


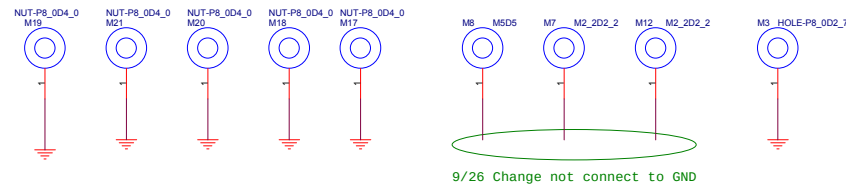
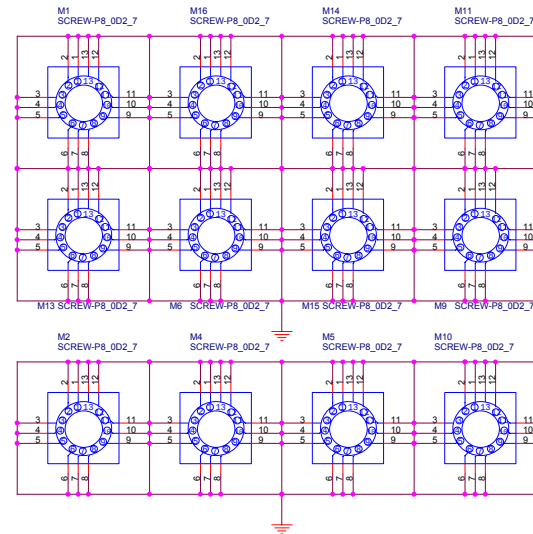
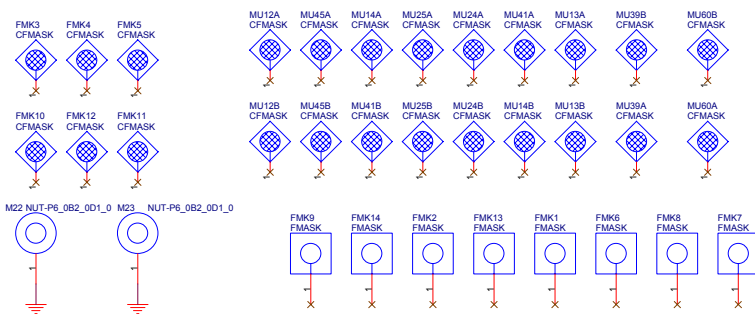
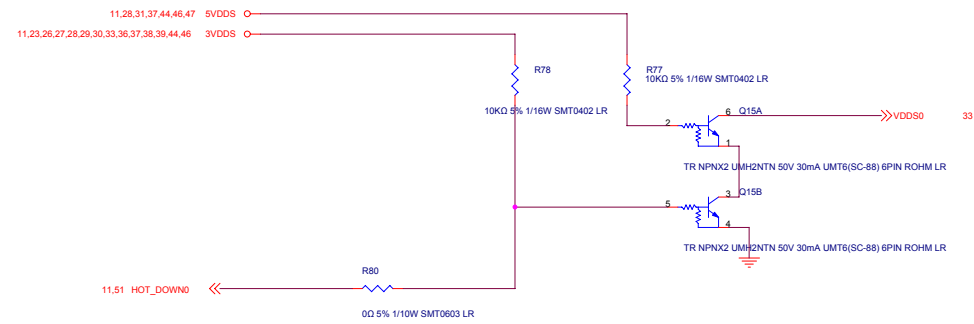
20-10161-00 SOCKET

RESUME
RESET

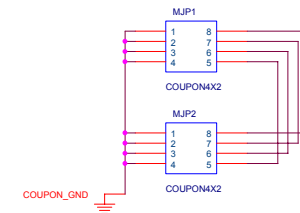


PCI RESET & PCI NON RESET

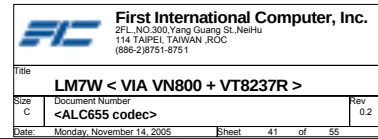


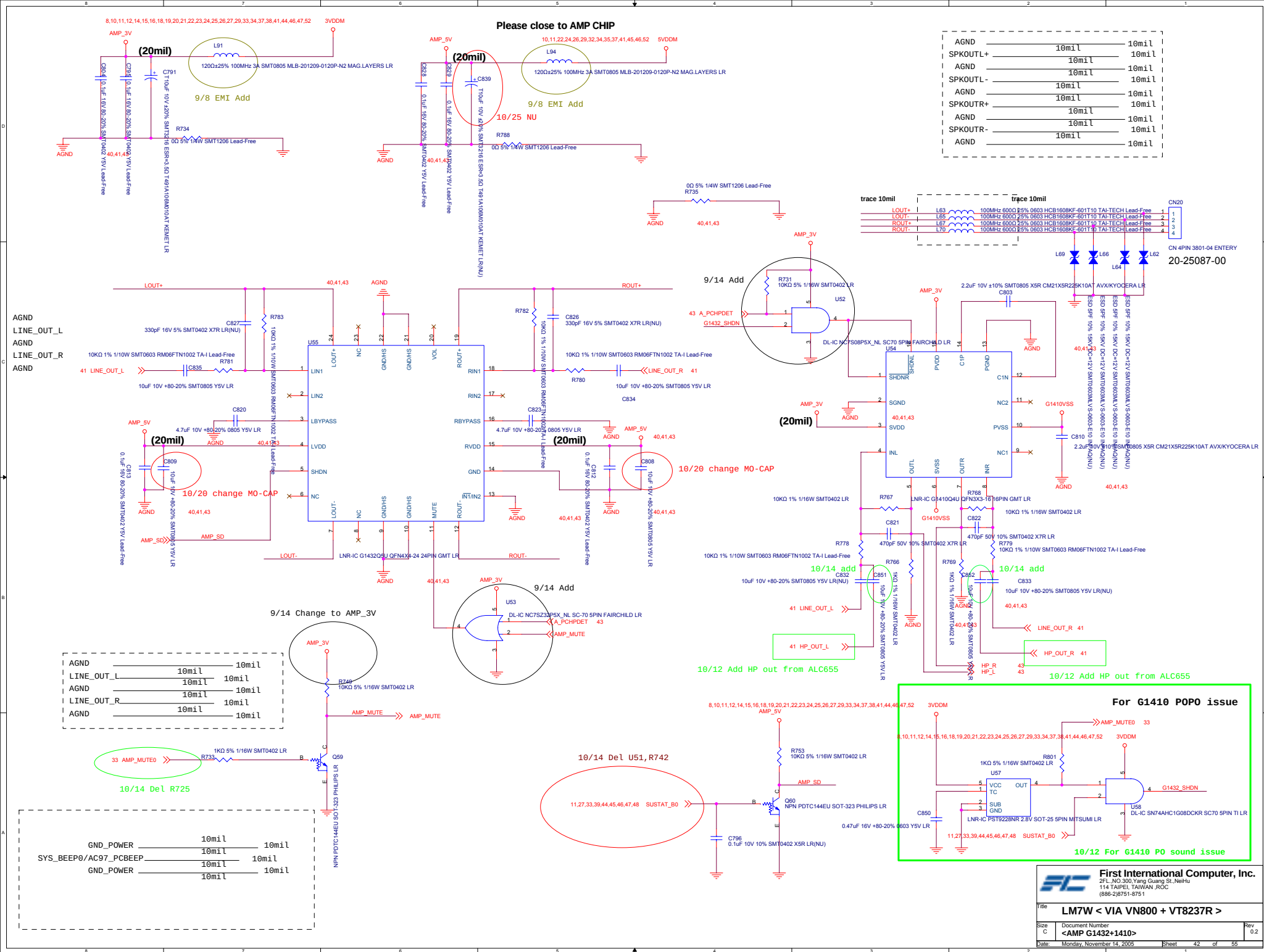


COUPON4X2

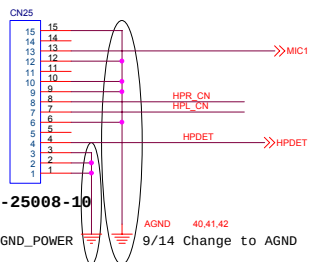


41,42,43

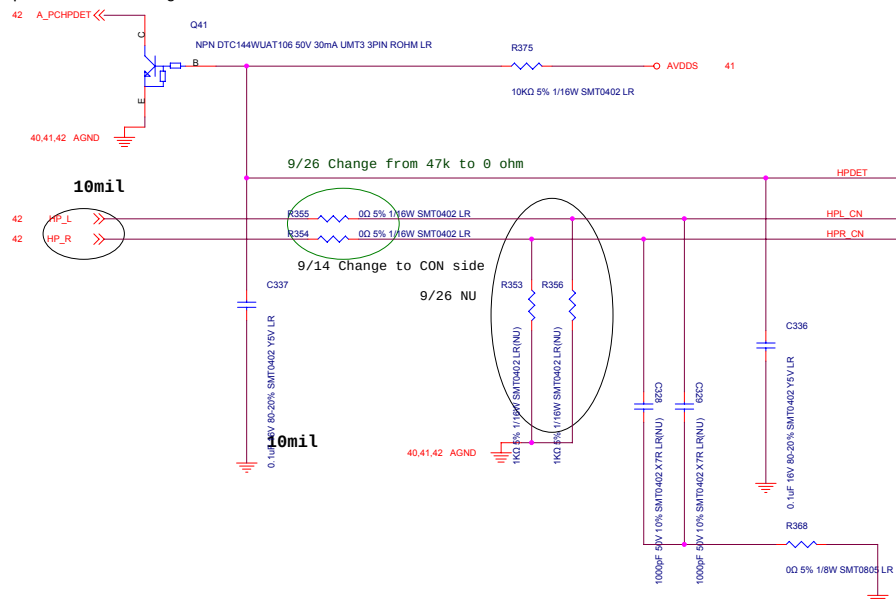




9/16 Cancel USB CON



Normal=LOW
Headphone insert=High

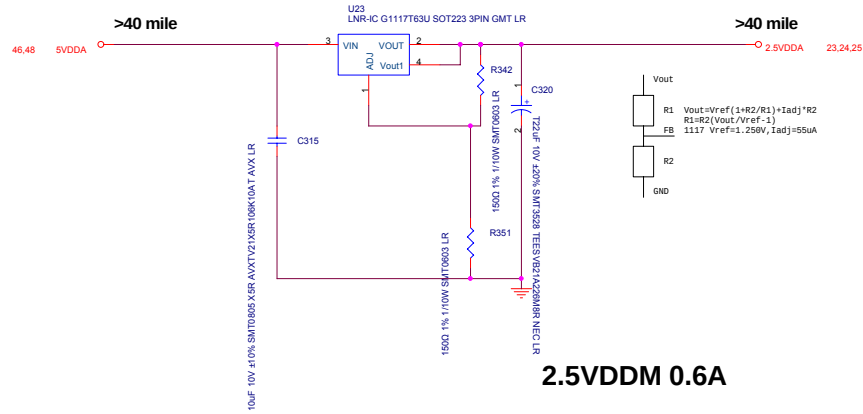


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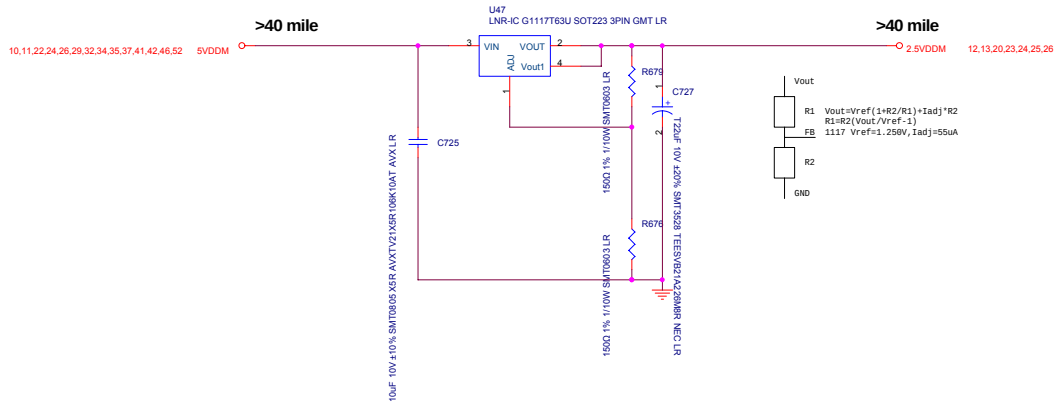
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9/12 Cancel

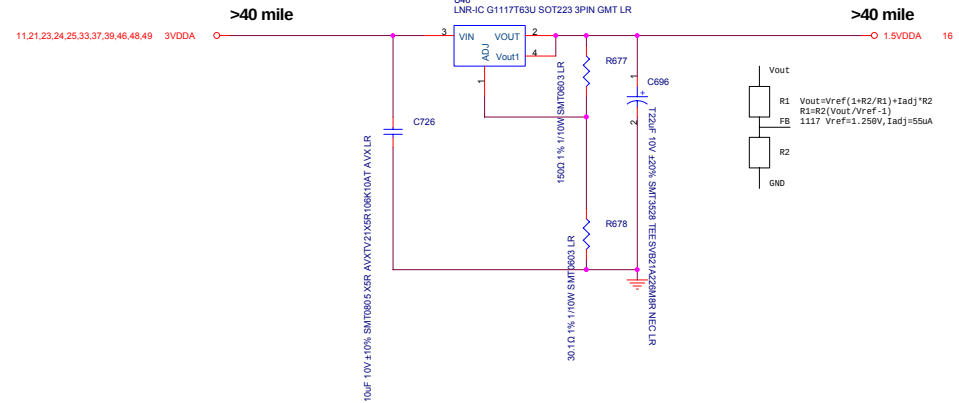
2.5VDDA 0.6A



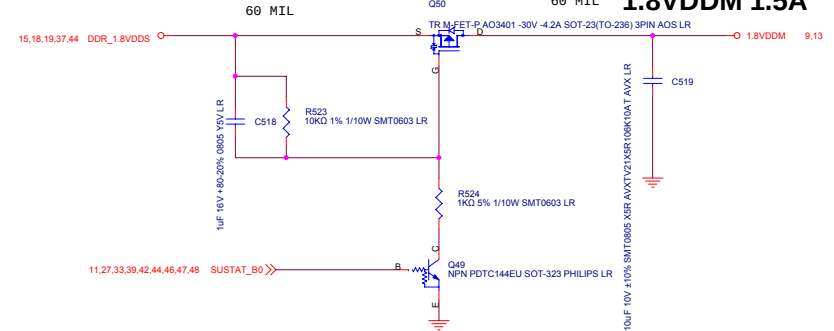
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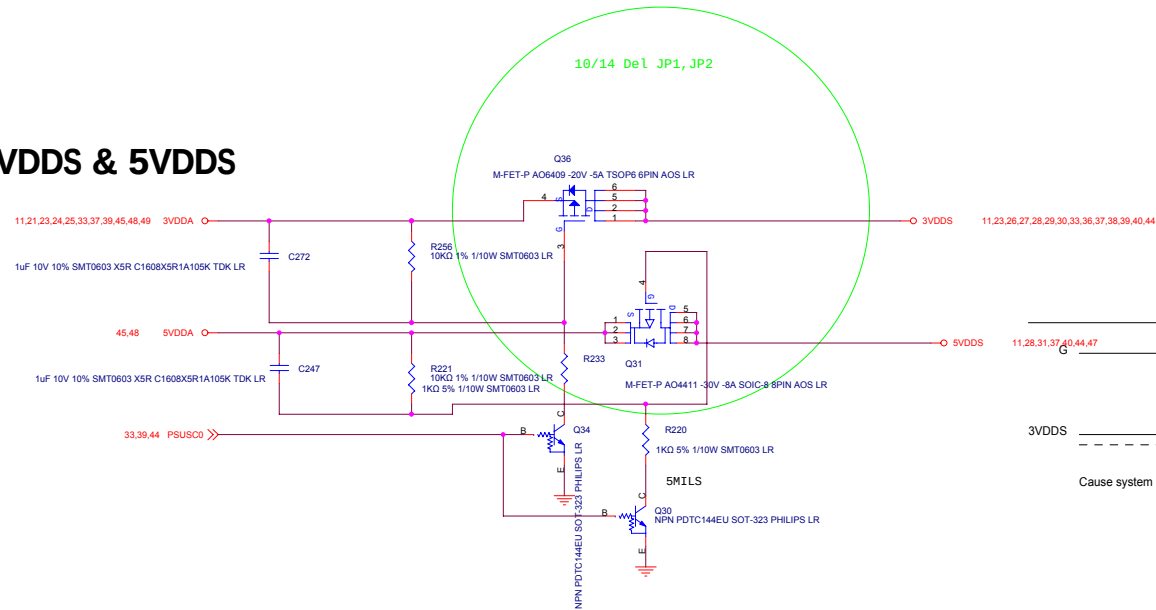
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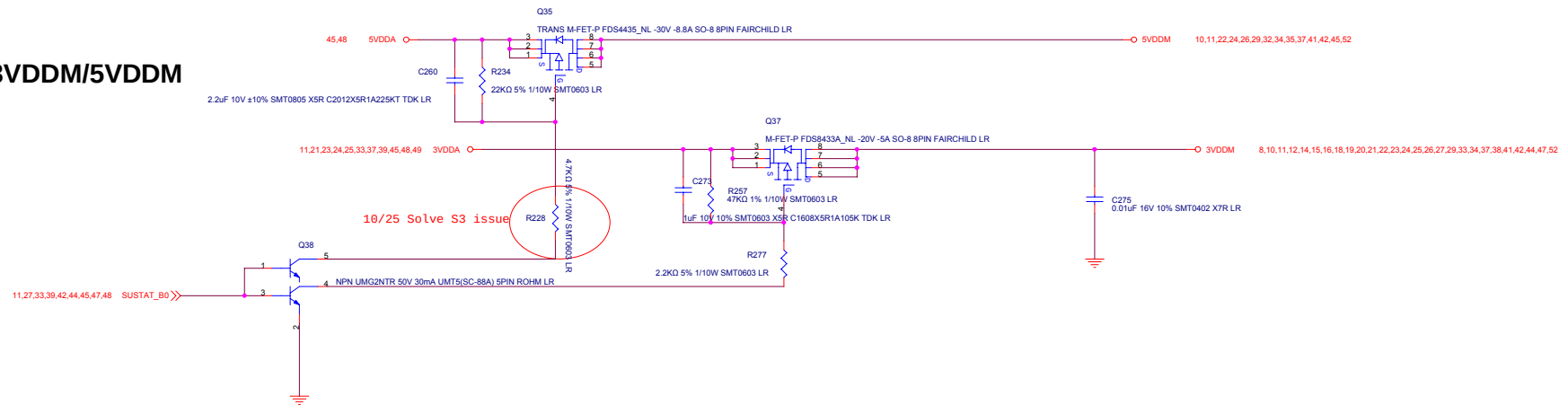
1.8VDDM 1.5A



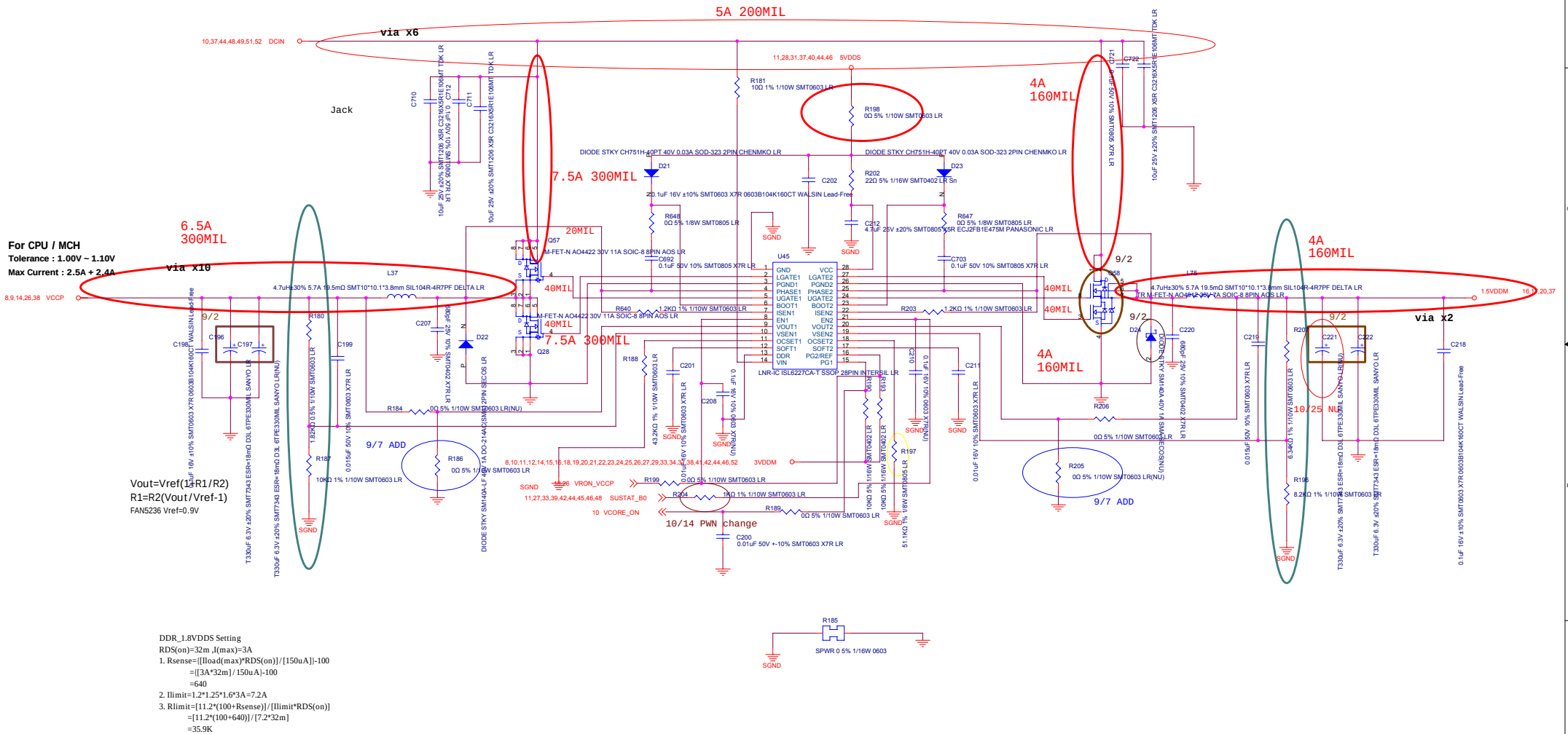
3VDDS & 5VDDS

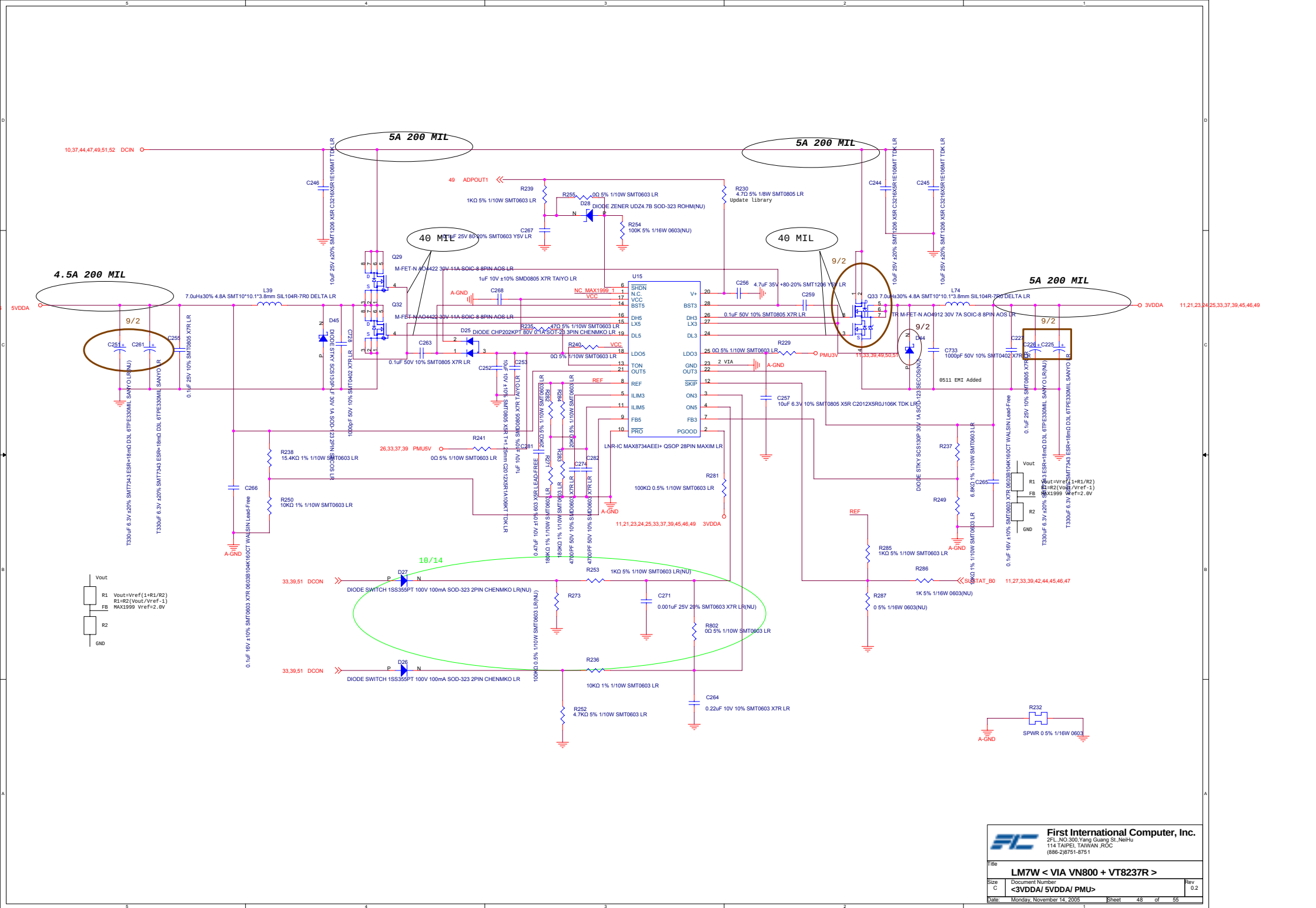


3VDDM/5VDDM



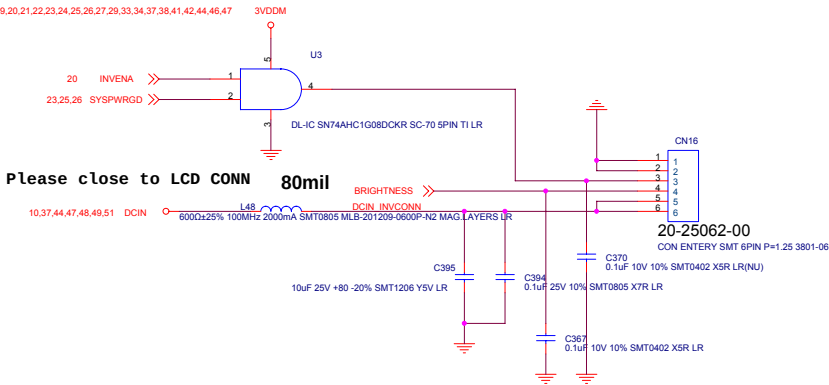
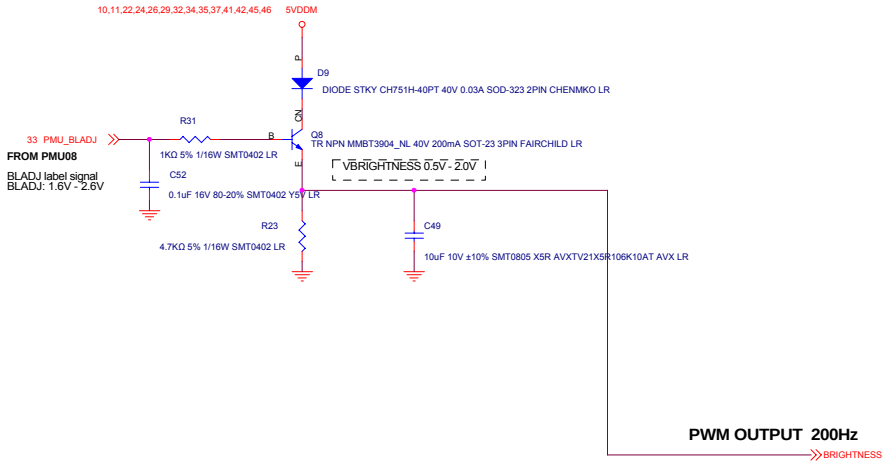
VCCP,VORE_GMCH





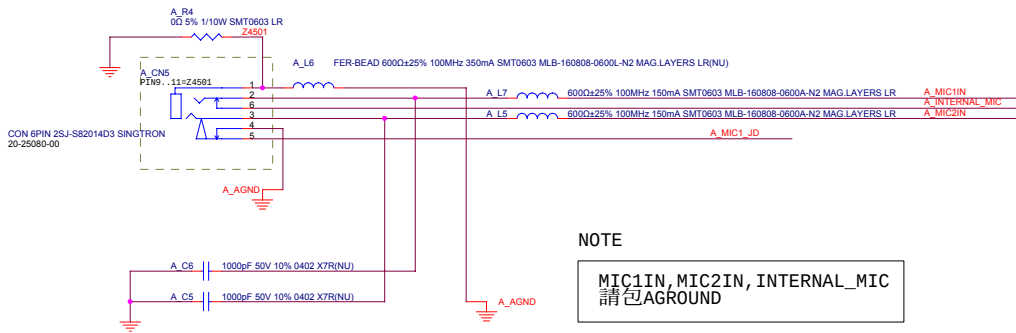
Inverter Control

LCD brightness control



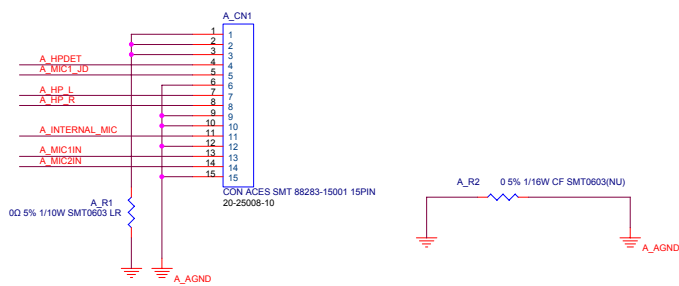
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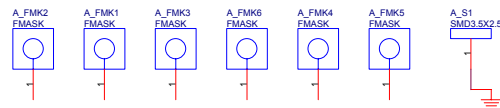
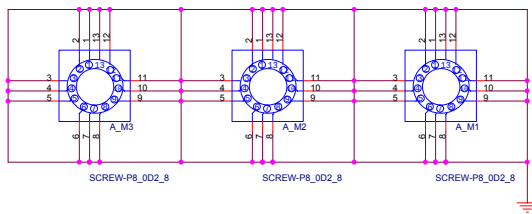
9/16 Cancel USB CON

9/16 Cancel USB CON



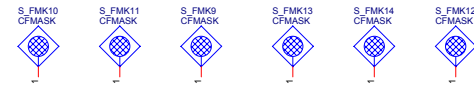
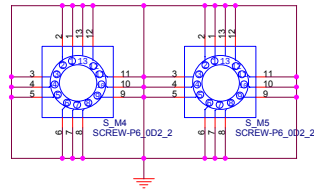
NOTE
HP_L, HP_R請包AGROUND

HEADPHONE JACK

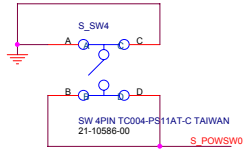


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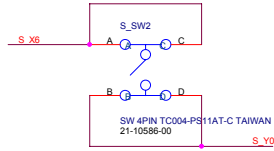
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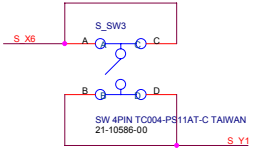
POWER SW



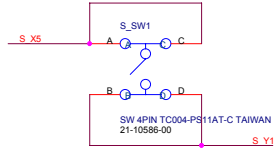
INTERNET SW



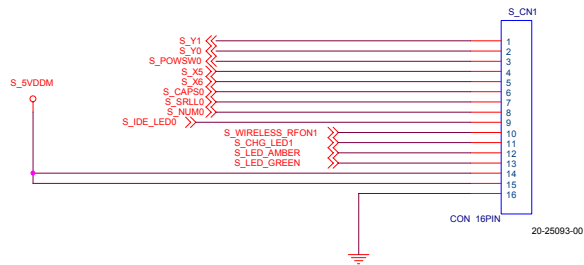
RF SW



SILEN SW

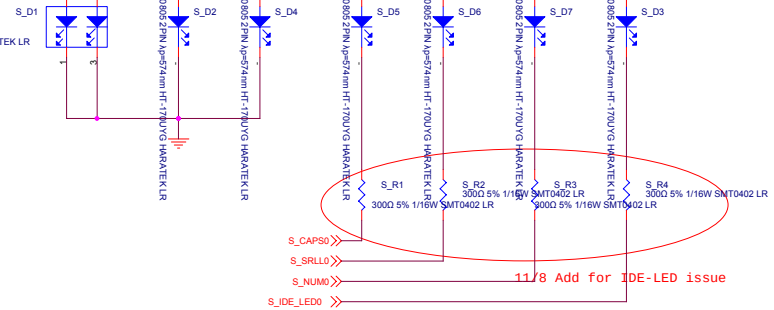


switch board con



S_WIRELESS_RFON1
S_CHG_LED1
S_LED_AMBER
S_LED_GREEN

LED AMBER/GREEN SMT0603 4PIN Ap=609/570mm HT-297DQ/GQ-DT HARVATEK LR



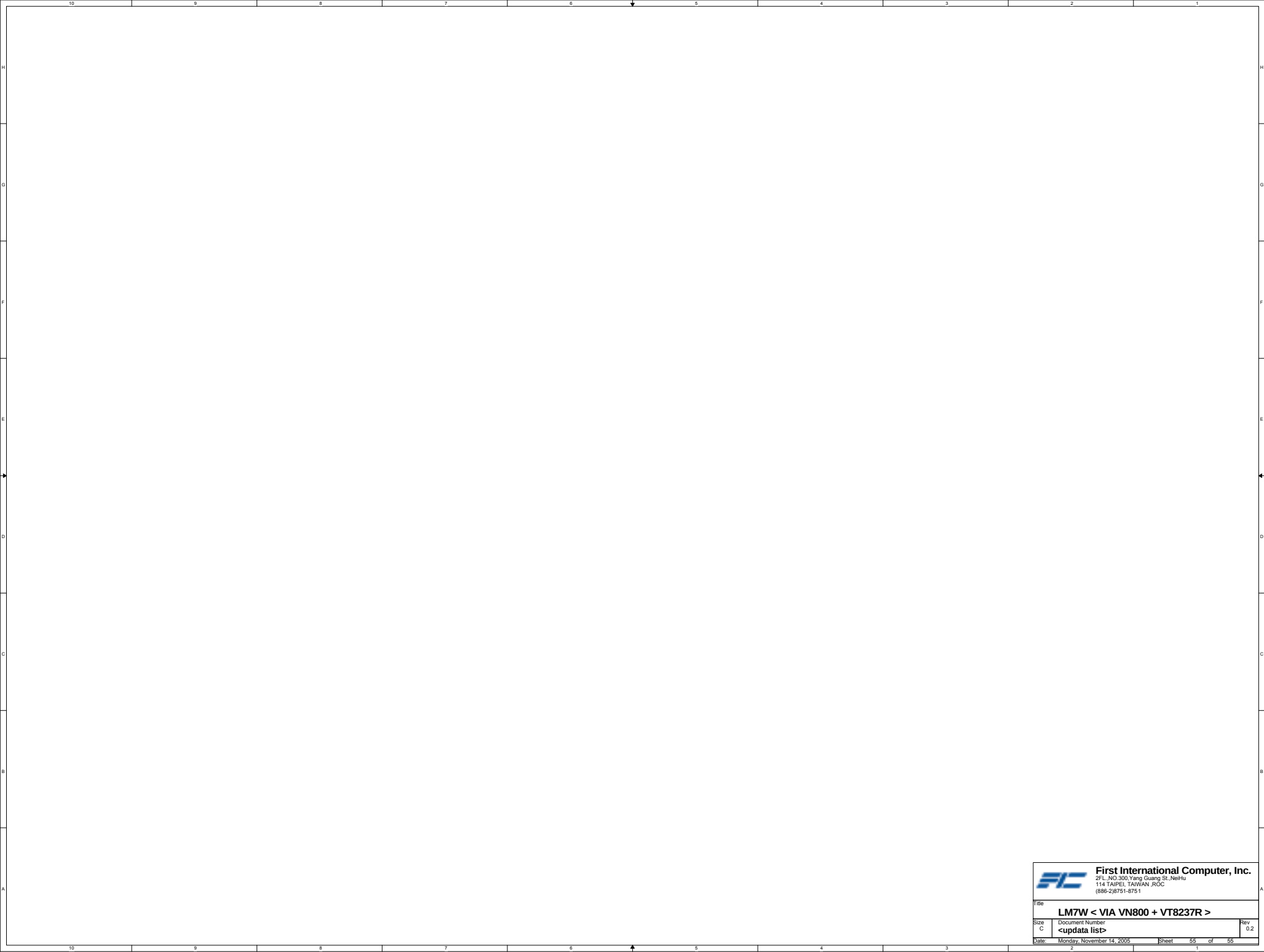
resistance on M/B

S_SVDDM



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<Switch transfer board>			
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Title LM7W < VIA VN800 + VT8237R >			
Size C	Document Number <update list>		Rev 0.2
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